Supplementary Information: Object-Conditioned HPID-Split: Deterministic Part-Identity Fusion for Editable Assets

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This supplement reports full diagnostics, reproducibility records, and auxiliary compatibility evidence. It does not expand the claims of the main paper.

S1. Scope, Evidence Map, and Data Isolation

Original benchmarks use annotation-derived object crops with automatic root and profile routing. Section S19.12 adds automatic primary-object processing of all 42 complete source images, paired with the original crops. Reference-root substitution is reported separately. Unseen-taxonomy discovery and occluded-input identity recovery remain unevaluated. Post-review extensions reuse existing cases rather than enlarge the independent test sets.

TABLE S1
Manifest and Leakage Audit

Split	Cases	SHA-256	Use
Development	65	3d16426e4389d59c9fc2f820e8efd3f12c06aa6b881c28e5914479cb1f42dbcb	inventory and review-rule development
Locked benchmark	226	633076503641e81bbc49e796146ff49de637212e9eb1bdc253b547f204f3f548	external/fine-ID outputs locked before reference access
Original group holdout	42	3d9d0191646fc2724f47976513a1c4f6a229c9f87866e3edc09527efd7205d33	local frozen protocol; metadata-aware eligibility; reference-blinded inference
PACO test annotation	226 + 42 selected	8d061428b86dc9a0e5c206f330ed579c61319c7d6884402e540ac33faed3d2a6	license remains with PACO/LVIS

The PACO taxonomy and selection metadata were shared. Reference-mask geometry and part labels were not supplied to inference and were loaded by the evaluator only for scoring.

TABLE S2
Cross-Category Benchmark Composition

Domain	n	Share
container	72	31.9%
daily_object	40	17.7%
device	59	26.1%
furniture	16	7.1%
tool_prop	31	13.7%
vehicle	8	3.5%

S2. Original Editable-Group Holdout

The release commit, configuration, model artifacts, domain quotas, and 42 case slots were frozen before materialization. Selection was annotation-metadata-aware: PACO test object category, annotated distinct-part count (at least two), and a minimum object-box side of 64 pixels were used for eligibility and quota filling. The rule did not use reference-mask geometry, model output, or mask quality. The selection environment could access the annotation dataset, but the operator did not inspect reference masks, and no reference mask or part label was supplied to inference or regrouping. All inference attempts finished before the evaluator loaded references for scoring. Public-link failures triggered the prespecified next-case integrity rule rather than case removal or rerunning.

TABLE S3
Original-Holdout Integrity Audit

Audit item	Result
Release commit	be543003632fa739d0b63e9acbffb8d91d99b84a
Protocol status	prespecified and locally frozen; not independently registered
Evidence archive commit	c45f9ce3abaecb43e814e86e3a3bb46173fc08a7
Unique objects / source images	42 / 42
Overlap with prior materialized data	0 objects; 0 source images
Inference attempts	42 cases; one attempt each
Reference access	reference masks and part labels loaded by evaluator after all 42 inference runs
Integrity fallbacks	15 total; 4 same-domain cross-category
Dimension normalization	30 images; aspect-ratio error <=1%
Valid packages	42/42
Process-isolated determinism	252/252 core files; 0 mismatch

The determinism count is 42 cases x six core files across two process-isolated regroup runs under Python hash seeds 20260825 and 20260826.

TABLE S4
Original-Holdout Aggregate Results

Metric	Mean	95% bootstrap CI	Draws
Precision@0.25	0.4023	[0.3195, 0.4889]	10000
Recall@0.25	0.4269	[0.3417, 0.5122]	10000
Group F1@0.25	0.3951	[0.3160, 0.4757]	10000
Matched IoU	0.4563	[0.3852, 0.5240]	10000
Boundary F1@0.25	0.5161	[0.4347, 0.5920]	10000
Semantic recall	0.2819	[0.2031, 0.3652]	10000
Oversegmentation	1.1862	[1.0219, 1.3597]	10000
Group F1@0.50	0.2417	[0.1618, 0.3303]	10000
Group F1@0.75	0.1035	[0.0435, 0.1777]	10000
Mean Group F1 (0.25:0.05:0.75)	0.2350	[0.1653, 0.3136]	10000

Group metrics here are the original frozen results. The original 42-object holdout uses automatic roots on annotation-derived crops, not reference roots. Section S19 adds a post-review fine-ID root substitution diagnostic without replacing these group results. Section S19.12 separately reports the full-image Group contrast for these same 42 targets.

TABLE S4A
Complete 42-Case Holdout Results

Case ID	Domain	Category	F1@0.25	F1@0.50	F1@0.75	Mean F1	IoU	B-F1@0.25	Sem. R
belt__u01	daily object	belt	0.4000	0.0000	0.0000	0.0727	0.3044	0.4577	0.0000
bench__u01	furniture	bench	0.2353	0.2353	0.1176	0.1818	0.7183	0.7744	0.4000
bicycle__u01	vehicle	bicycle	0.2353	0.0000	0.0000	0.0642	0.3962	0.4674	0.0833
bicycle__u02	vehicle	bicycle	0.4444	0.1111	0.1111	0.2121	0.4920	0.6464	0.1667
book__u01	daily object	book	0.3333	0.3333	0.0000	0.1818	0.5076	0.6388	0.5000
bottle__u01	container	bottle	0.5000	0.1667	0.0000	0.2121	0.4627	0.4740	0.2857
bowl__u01	container	bowl	0.4000	0.2000	0.0000	0.2182	0.5328	0.5346	0.4000
box__u01	container	box	0.4444	0.4444	0.2222	0.3434	0.7308	0.5846	0.0000
calculator__u01	device	cellular telephone	0.5000	0.5000	0.2500	0.4091	0.6865	0.5382	0.5000
can__u01	container	can	0.7273	0.3636	0.1818	0.4463	0.5754	0.7615	0.8000
car_automobile__u01	vehicle	car (automobile)	0.4516	0.1935	0.0000	0.1877	0.4428	0.6975	0.3750
car_automobile__u02	vehicle	car (automobile)	0.2400	0.1600	0.0800	0.1527	0.5714	0.7201	0.0667
cellular_telephone__u01	device	cellular telephone	0.5455	0.3636	0.1818	0.3802	0.5995	0.2463	0.5000
chair__u01	furniture	chair	0.2000	0.2000	0.0000	0.1091	0.5406	0.7250	0.0000
clock__u01	device	clock	0.3636	0.0000	0.0000	0.1322	0.4204	0.6815	0.6667
cup__u01	container	cup	0.2000	0.2000	0.0000	0.1273	0.5540	0.5372	0.2500
glass_drink_container__u01	container	glass (drink container)	0.2222	0.2222	0.0000	0.1616	0.6189	0.4498	0.2500
guitar__u01	tool prop	guitar	0.5714	0.2857	0.0000	0.2857	0.5006	0.6420	0.4000
handbag__u01	container	handbag	0.4444	0.0000	0.0000	0.0808	0.3057	0.4181	0.5000
hat__u01	daily object	hat	0.5000	0.5000	0.0000	0.3636	0.6156	0.9378	0.5000
helmet__u01	daily object	helmet	0.3333	0.0000	0.0000	0.1212	0.4229	0.7776	0.3333
jar__u01	container	jar	0.3636	0.0000	0.0000	0.1322	0.4179	0.6043	0.2000
knife__u01	tool prop	knife	1.0000	1.0000	1.0000	1.0000	0.8549	0.7085	1.0000
ladder__u01	tool prop	ladder	0.5714	0.5714	0.0000	0.3117	0.5275	0.5793	0.0000
laptop_computer__u01	device	laptop computer	0.4706	0.2353	0.1176	0.2995	0.6157	0.5181	0.4286
mug__u01	container	mug	0.7273	0.7273	0.3636	0.5785	0.6796	0.6036	0.6667
pan_for_cooking__u01	tool prop	pan (for cooking)	0.2857	0.0000	0.0000	0.0519	0.3105	0.4818	0.0000
pencil__u01	daily object	pencil	0.5000	0.5000	0.0000	0.3182	0.5953	0.8936	0.0000

Case ID	Domain	Category	F1@0.25	F1@0.50	F1@0.75	Mean F1	IoU	B-F1@0.25	Sem. R
plastic_bag__u01	container	plastic bag	0.2500	0.2500	0.0000	0.1364	0.5301	0.7252	0.3333
pliers__u01	tool prop	pliers	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
remote_control__u01	device	remote control	0.2000	0.0000	0.0000	0.0545	0.3559	0.4349	0.3333
scarf__u01	daily object	scarf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
scissors__u01	tool prop	scissors	0.8889	0.2222	0.2222	0.3434	0.4403	0.5199	0.2500
screwdriver__u01	tool prop	knife	1.0000	1.0000	1.0000	1.0000	0.8877	0.9563	0.0000
shoe__u01	daily object	shoe	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
slipper_footwear__u01	daily object	book	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
stool__u01	furniture	stool	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
table__u01	furniture	table	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
telephone__u01	device	telephone	0.3333	0.1667	0.0000	0.1212	0.4306	0.3943	0.2500
television_set__u01	device	television set	0.2500	0.0000	0.0000	0.0682	0.3795	0.2661	0.0000
watch__u01	device	watch	0.4615	0.0000	0.0000	0.0559	0.2996	0.4710	0.4000
wrench__u01	tool prop	knife	1.0000	1.0000	0.5000	0.9545	0.8399	0.8070	1.0000

All 42 completed cases are included. F1@0.50 and F1@0.75 are independently matched at their stated thresholds; Mean F1 averages 0.25:0.05:0.75. No case was removed after scoring.

TABLE S4B
Complete Holdout Fallback Audit

Slot	Planned object	Final object	Fallback	Same cat.	Failure reason
cellular_telephone__u01	cellular_telephone|obj=65705|img=444746	cellular_telephone|obj=65705|img=444746	false	true	none
laptop_computer__u01	laptop_computer|obj=187692|img=578775	laptop_computer|obj=187692|img=578775	false	true	none
television_set__u01	television_set|obj=92753|img=347389	television_set|obj=92753|img=347389	false	true	none
remote_control__u01	remote_control|obj=17499|img=229111	remote_control|obj=17499|img=229111	false	true	none
calculator__u01	calculator|obj=38531|img=561366	cellular_telephone|obj=70677|img=104982	true	false	2 rejected candidate(s): HTTP 404; TLS certificate mismatch
telephone__u01	telephone|obj=98939|img=70809	telephone|obj=102790|img=523919	true	true	1 rejected candidate(s): HTTP 404; TLS certificate mismatch
clock__u01	clock|obj=57760|img=72989	clock|obj=57760|img=72989	false	true	none
watch__u01	watch|obj=148791|img=195188	watch|obj=148791|img=195188	false	true	none

Slot	Planned object	Final object	Fallback	Same cat.	Failure reason
chair__u01	chair|obj=238100|img=129407	chair|obj=95920|img=73549	true	true	1 rejected candidate(s): HTTP 404; TLS certificate mismatch
bench__u01	bench|obj=146291|img=545181	bench|obj=146291|img=545181	false	true	none
stool__u01	stool|obj=104289|img=441411	stool|obj=104289|img=441411	false	true	none
table__u01	table|obj=116675|img=68162	table|obj=116675|img=68162	false	true	none
bottle__u01	bottle|obj=20822|img=460841	bottle|obj=20822|img=460841	false	true	none
jar__u01	jar|obj=71734|img=223444	jar|obj=71734|img=223444	false	true	none
can__u01	can|obj=105328|img=172012	can|obj=105328|img=172012	false	true	none
cup__u01	cup|obj=87909|img=163138	cup|obj=87909|img=163138	false	true	none
mug__u01	mug|obj=150618|img=249130	mug|obj=15210|img=2592	true	true	1 rejected candidate(s): HTTP 404; TLS certificate mismatch
glass_drink_container__u01	glass_(drink_container)|obj=40606|img=85089	glass_(drink_container)|obj=40606|img=85089	false	true	none
bowl__u01	bowl|obj=244436|img=319996	bowl|obj=244436|img=319996	false	true	none
handbag__u01	handbag|obj=51856|img=467742	handbag|obj=53054|img=475995	true	true	2 rejected candidate(s): HTTP 404; TLS certificate mismatch
plastic_bag__u01	plastic_bag|obj=93155|img=283131	plastic_bag|obj=93155|img=283131	false	true	none
box__u01	box|obj=77503|img=5754	box|obj=182311|img=291898	true	true	2 rejected candidate(s): HTTP 404; TLS certificate mismatch
car_automobile__u01	car_(automobile)|obj=45908|img=67616	car_(automobile)|obj=45908|img=67616	false	true	none
bicycle__u01	bicycle|obj=220953|img=358242	bicycle|obj=220953|img=358242	false	true	none
car_automobile__u02	car_(automobile)|obj=218007|img=246750	car_(automobile)|obj=218007|img=246750	false	true	none
bicycle__u02	bicycle|obj=133274|img=187509	bicycle|obj=52568|img=453138	true	true	2 rejected candidate(s): HTTP 404; TLS certificate mismatch
book__u01	book|obj=76306|img=159561	book|obj=208173|img=231351	true	true	1 rejected candidate(s): HTTP 404; TLS certificate mismatch

Slot	Planned object	Final object	Fallback	Same cat.	Failure reason
pencil__u01	pencil|obj=190713|img=310463	pencil|obj=118123|img=287285	true	true	1 rejected candidate(s): TLS certificate mismatch
belt__u01	belt|obj=225859|img=419489	belt|obj=225859|img=419489	false	true	none
hat__u01	hat|obj=8203|img=19924	hat|obj=224917|img=468519	true	true	1 rejected candidate(s): TLS certificate mismatch
helmet__u01	helmet|obj=154250|img=221343	helmet|obj=154250|img=221343	false	true	none
shoe__u01	shoe|obj=30158|img=35197	shoe|obj=30158|img=35197	false	true	none
slipper_footwear__u01	slipper_(footwear)|obj=9366|img=18150	book|obj=150053|img=281774	true	false	3 rejected candidate(s): TLS certificate mismatch
scarf__u01	scarf|obj=63479|img=539259	scarf|obj=73790|img=434689	true	true	2 rejected candidate(s): HTTP 404; TLS certificate mismatch
screwdriver__u01	screwdriver|obj=232603|img=195288	knife|obj=64448|img=537081	true	false	1 rejected candidate(s): HTTP 404; TLS certificate mismatch
wrench__u01	wrench|obj=105318|img=24136	knife|obj=153190|img=68344	true	false	1 rejected candidate(s): TLS certificate mismatch
knife__u01	knife|obj=210603|img=370728	knife|obj=210603|img=370728	false	true	none
scissors__u01	scissors|obj=759|img=351096	scissors|obj=759|img=351096	false	true	none
pliers__u01	pliers|obj=161597|img=528119	pliers|obj=161597|img=528119	false	true	none
ladder__u01	ladder|obj=148550|img=327988	ladder|obj=148550|img=327988	false	true	none
guitar__u01	guitar|obj=225848|img=302715	guitar|obj=192151|img=92041	true	true	1 rejected candidate(s): HTTP 404; TLS certificate mismatch
pan_for_cooking__u01	pan_(for_cooking)|obj=115950|img=143510	pan_(for_cooking)|obj=115950|img=143510	false	true	none

All 42 slots are shown. The complete JSON preserves every rejected candidate and raw source-integrity record; fallback occurred before inference and never used predictions or mask quality.

TABLE S4C
Complete Holdout Resize and Coordinate Audit

Slot	Resize	Downloaded	Annotation	Aspect error	Interpolation
cellular_telephone__u01	true	640x480	500x375	0.00000000	PIL Lanczos
laptop_computer__u01	true	639x359	640x359	0.00156250	PIL Lanczos
television_set__u01	true	640x480	500x375	0.00000000	PIL Lanczos
remote_control__u01	true	480x640	375x500	0.00000000	PIL Lanczos
calculator__u01	true	639x359	500x281	0.00032869	PIL Lanczos
telephone__u01	false	unchanged	native	0	none
clock__u01	true	445x639	446x640	0.00068072	PIL Lanczos
watch__u01	true	640x213	500x166	0.00244131	PIL Lanczos
chair__u01	false	unchanged	native	0	none
bench__u01	true	640x480	500x375	0.00000000	PIL Lanczos
stool__u01	true	639x428	640x428	0.00156250	PIL Lanczos
table__u01	true	639x426	640x426	0.00156250	PIL Lanczos
bottle__u01	true	640x480	500x375	0.00000000	PIL Lanczos
jar__u01	true	639x368	500x288	0.00017391	PIL Lanczos
can__u01	true	640x480	500x375	0.00000000	PIL Lanczos
cup__u01	true	640x480	500x375	0.00000000	PIL Lanczos
mug__u01	true	639x365	640x366	0.00117295	PIL Lanczos
glass_drink_container__u01	true	639x428	640x428	0.00156250	PIL Lanczos
bowl__u01	true	532x639	416x500	0.00066209	PIL Lanczos
handbag__u01	false	unchanged	native	0	none
plastic_bag__u01	true	640x480	500x375	0.00000000	PIL Lanczos
box__u01	false	unchanged	native	0	none
car_automobile__u01	false	unchanged	native	0	none
bicycle__u01	true	639x511	640x511	0.00156250	PIL Lanczos
car_automobile__u02	false	unchanged	native	0	none
bicycle__u02	true	640x480	500x375	0.00000000	PIL Lanczos
book__u01	true	432x576	375x500	0.00000000	PIL Lanczos
pencil__u01	true	640x426	500x333	0.00056338	PIL Lanczos
belt__u01	false	unchanged	native	0	none
hat__u01	true	640x480	500x375	0.00000000	PIL Lanczos
helmet__u01	true	640x427	640x426	0.00234192	PIL Lanczos
shoe__u01	true	428x639	429x640	0.00076970	PIL Lanczos
slipper_footwear__u01	true	640x480	500x375	0.00000000	PIL Lanczos
scarf__u01	true	469x639	367x500	0.00005543	PIL Lanczos

Slot	Resize	Downloaded	Annotation	Aspect error	Interpolation
screwdriver__u01	false	unchanged	native	0	none
wrench__u01	false	unchanged	native	0	none
knife__u01	false	unchanged	native	0	none
scissors__u01	true	640x480	500x375	0.00000000	PIL Lanczos
pliers__u01	true	427x639	428x640	0.00077516	PIL Lanczos
ladder__u01	false	unchanged	native	0	none
guitar__u01	false	unchanged	native	0	none
pan_for_cooking__u01	true	639x425	640x425	0.00156250	PIL Lanczos

RGB normalization uses PIL Lanczos only after <=1% aspect-ratio agreement. Object and part masks are decoded at annotation dimensions and cropped with the same oracle object box, preserving one coordinate system.

TABLE S5
Original Holdout by Domain

Domain	n	P@0.25	R@0.25	F1@0.25	IoU	B-F1@0.25	Sem. R	Overseg.
container	10	0.4167	0.4529	0.4279	0.5408	0.5693	0.3686	1.1864
daily object	8	0.2396	0.2917	0.2583	0.3057	0.4632	0.1667	1.1806
device	8	0.3350	0.5152	0.3906	0.4735	0.4438	0.3848	1.6723
furniture	4	0.1250	0.1357	0.1088	0.3147	0.3749	0.1000	1.2238
tool prop	8	0.7250	0.6583	0.6647	0.5452	0.5868	0.3312	0.9646
vehicle	4	0.4583	0.2844	0.3428	0.4756	0.6329	0.1729	0.6302

Domain rows are descriptive; furniture and vehicle each contain four cases. No domain-level confidence or superiority claim is made.

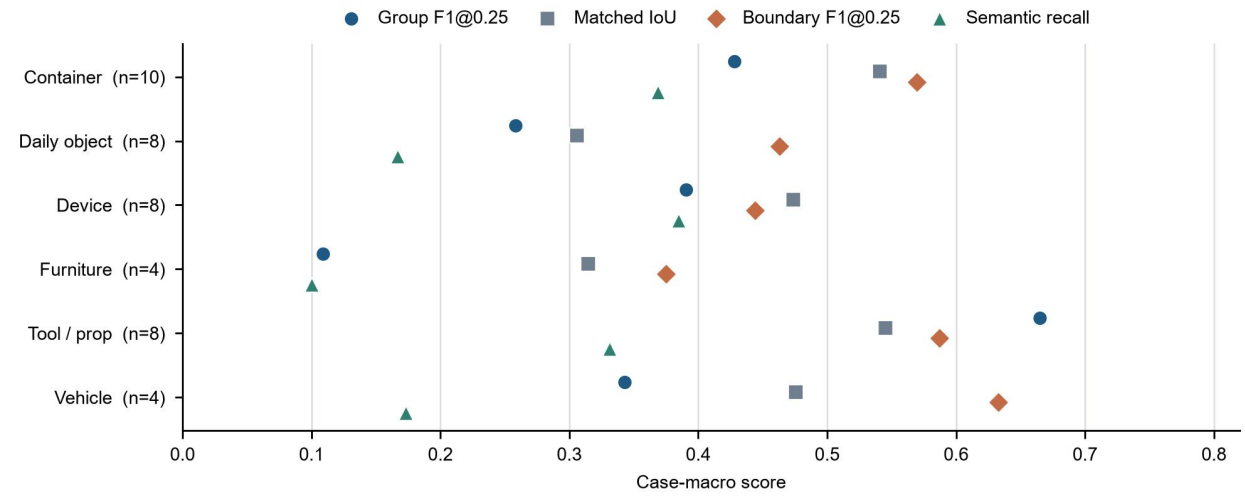


Fig. S1. All 42 original holdout cases summarized by domain. Points are case-macro domain means; no case was removed. Small-domain rows are descriptive, and aggregate bootstrap intervals are reported in Table S4.

S3. Metrics and Matching

Predicted and reference masks are assigned one-to-one by Hungarian maximization of pairwise mask IoU. At each threshold, accepted assignments form the true-positive count; unmatched predictions remain in the precision denominator and unmatched references remain in the recall denominator. Precision, recall, and their harmonic-mean F1 are computed per case and then macro-averaged independently. Mean F1 averages independently matched results over 0.25:0.05:0.75. Semantic F1@0.25 first normalizes names by lowercasing, replacing nonalphanumeric runs with underscores, and applying the frozen taxonomy aliases; Hungarian matching is then restricted to equal normalized names and masks accepted at IoU 0.25, while unlabeled predictions remain in the precision denominator. Boundary pixels are a mask XOR its 3 x 3 erosion; Boundary F1 uses a three-pixel square-dilation tolerance on pairs accepted at the corresponding IoU threshold. Object IoU compares the union of scored predicted part masks with the reference object mask. Root-foreground IoU additionally includes root/residual

ownership; it is distinct from the IoU of the input root mask. Overlap excess is the root-normalized count of extra owners, unassigned root is the root-normalized uncovered count, and oversegmentation is predicted-part count divided by reference-part count. Holdout aggregate intervals use 10,000 case-level percentile bootstrap draws with seed 20260825. Original locked method and A0-A3 differences use 20,000 case-paired percentile draws with seed 20260822 and resample the same object-case indices for both methods or variants. The completion/root audits and new post-review diagnostics use their separately specified 10,000-draw settings.

TABLE S6
Metric Interpretation

Metric	Interpretation
Part F1@0.25	approximate editable ownership
Part F1@0.50 / @0.75	increasingly strict localization
Semantic F1@0.25	IoU 0.25 overlap plus normalized part-name agreement
Boundary F1@threshold	three-pixel-tolerant boundary agreement on masks accepted at that IoU threshold
Object IoU	union of scored predicted part masks compared with the reference object mask
Oversegmentation	predicted/reference part-count ratio

TABLE S6A
Core Release Parameters for Equations (2)-(3)

Term	Frozen value/range	Operational definition
w_i	[0.01, 0.995]	Clipped effective proposal strength from the stored ranking score, reliability and agreement factor; applied once to soft membership before the within-class/family maximum
Parent attenuation	1 or $0.12^{0.65}$ = 0.2520	When compatible parent support exists, evidence is unchanged inside its dilated region and attenuated outside
$d_k(\mathrm{x})$	{0,1}; direct support > 0.38	Detail-class indicator; qualifying pixels receive a +0.08 score bonus
Direct-gate margin	0.82	Qualifying detail scores must be at least this fraction of the all-class maximum; direct and dilated-parent support must also meet their thresholds
$o_k(\mathrm{x})$	{0,1}	Indicator for eligible broad-host suppression
Host retention / effective penalty	0.28 / 0.72	At marked pixels a broad host retains 28% of $E_k(\mathrm{x})$, the evidence after parent, residual and child-floor updates
Child evidence floor	0.90 x host evidence	Eligible children receive at least 90% of corresponding host evidence before host suppression

Equations (2) and (3) distinguish proposal, semantic-class and family indices. Within-family maxima include each candidate's effective weight exactly once. Before the displayed host-suppression/detail-bonus score, the implementation applies parent attenuation (one inside compatible support and 0.12^0.65 outside), parent-residual multiplication by one minus 0.88 times the strongest child evidence, and eligible child evidence floors of 0.90 times the corresponding host evidence. The background score is the clipped complement of maximum parent support ([0.04, 0.98]), capped at 0.01 within accepted roots. The all-class argmax includes background; the direct-detail gate replaces it only at pixels with qualifying detail labels. Cleanup, root restoration and fine-instance construction follow. The optional boundary-ownership branch is disabled in the release. Section S19.7 records the actual intermediate states and the registries retain all defaults.

TABLE S6B
Recorded Source-Family Reliabilities in the 42-Case Holdout

Source family	n	Recorded raw reliability	Effective w_i range
Conditional-part direct	65	0.7840–0.9181	0.5516–0.8779
Grounded-SAM2 root/retrieved	52	0.6628–0.9976	0.4065–0.9506

Source family	n	Recorded raw reliability	Effective w_i range
Profile-dense supplement	9	0.4215–0.8470	0.2721–0.7916
Grounded-SAM2 profile-refine	95	0.6688–1.1632	0.3898–0.7548
Appearance contour	8	0.6200–0.6600	0.4755–0.5408
Repetitive physical detail	8	0.7600	0.5581–0.6329
Topology	6	0.8200	0.4805–0.6368
SAM2 automatic masks	206	0.5819–0.7400	0.4537–0.6437

For each proposal, s is the final stored source-provided proposal score, after any recorded generation, relabeling, and appearance-graph transformations, and r is the stored source-reliability multiplier. The fusion rule is w = clip((0.45 + 0.55 clip(s, 0, 1)) r a, 0.01, 0.995). Broad scene fallback layers receive an additional factor of 0.72 followed by the same clipping. The optional agreement factor a is exactly 1 in the released configuration. The multiplier r is a source/profile heuristic and can exceed 1; only the final effective strength is clipped. Within each semantic-class/source-family pair, the strongest pixelwise weighted support is retained; noisy-OR combines those family surfaces. Explicit family metadata is used when available; otherwise the last source-path component is removed and Grounding-DINO tiny/base naming is normalized. Profile-dense and profile-refine remain separate recorded families. These strengths are not statistically calibrated probabilities. Source definitions, provenance, annotation-conditional reliability diagnostics and the controlled consensus tests are in Section S19.9; parameter sensitivity is in Section S19.1.

S4. External Baselines by Domain

External predictors include SAM2, Grounded-SAM2, a CLIPSeg adaptation and the newly executed official VOC-trained PartCATSeg checkpoint. The PartCATSeg comparison uses all 42 original holdout crops and a fixed PACO vocabulary, with a documented semantic-to-instance adapter (S19.14). HOPS and KPS have no verified trained accuracy run in this study. The frozen 1,951-proposal experiment remains the controlled comparison of postprocessing alone.

TABLE S7
Executed Baselines by Domain

Method	Domain	n	F1@0.25	F1@0.50	B-F1@0.50	Obj. IoU	Overseg.
SAM2 raw	container	72	0.1648	0.1159	0.4058	0.3823	3.9144
SAM2 raw	daily_object	40	0.1089	0.0665	0.2228	0.2994	3.4339
SAM2 raw	device	59	0.1750	0.1250	0.5222	0.3525	3.0572
SAM2 raw	furniture	16	0.1847	0.1187	0.5062	0.2141	3.7421
SAM2 raw	tool_prop	31	0.1467	0.1148	0.5019	0.1456	3.8575
SAM2 raw	vehicle	8	0.1697	0.1162	0.7826	0.4586	1.7874
SAM2+NMS	container	72	0.1650	0.1093	0.3909	0.3310	3.0799
SAM2+NMS	daily_object	40	0.1115	0.0631	0.2115	0.2778	2.7218
SAM2+NMS	device	59	0.1830	0.1258	0.4754	0.3072	2.3390
SAM2+NMS	furniture	16	0.1935	0.1282	0.5059	0.2109	3.1147
SAM2+NMS	tool_prop	31	0.1395	0.1089	0.4997	0.1337	3.3239
SAM2+NMS	vehicle	8	0.1191	0.0898	0.4706	0.3681	1.3320
SAM2+max-own.	container	72	0.1605	0.1072	0.3763	0.3310	3.0799
SAM2+max-own.	daily_object	40	0.1073	0.0631	0.2118	0.2778	2.7218
SAM2+max-own.	device	59	0.1769	0.1220	0.4650	0.3072	2.3390
SAM2+max-own.	furniture	16	0.1935	0.1282	0.5059	0.2109	3.1147
SAM2+max-own.	tool_prop	31	0.1313	0.1057	0.4955	0.1337	3.3239

Method	Domain	n	F1@0.25	F1@0.50	B-F1@0.50	Obj. IoU	Overseg.
SAM2+max-own.	vehicle	8	0.1056	0.0831	0.3449	0.3681	1.3320
Grounded-SAM2	container	72	0.2304	0.1555	0.2431	0.3288	0.3630
Grounded-SAM2	daily_object	40	0.1917	0.1067	0.1475	0.3529	0.3862
Grounded-SAM2	device	59	0.2936	0.1993	0.4404	0.3623	0.4858
Grounded-SAM2	furniture	16	0.3165	0.1678	0.4091	0.2675	0.5269
Grounded-SAM2	tool_prop	31	0.2285	0.1832	0.3354	0.2845	0.3952
Grounded-SAM2	vehicle	8	0.3466	0.2156	0.7152	0.5012	0.4501
HPID-Split A3	container	72	0.3237	0.1271	0.2797	0.6353	1.3097
HPID-Split A3	daily_object	40	0.2737	0.0843	0.1987	0.4563	1.7224
HPID-Split A3	device	59	0.3392	0.1953	0.4583	0.5735	2.1404
HPID-Split A3	furniture	16	0.2807	0.1408	0.3686	0.3343	1.4534
HPID-Split A3	tool_prop	31	0.2963	0.2047	0.3768	0.3278	0.9745
HPID-Split A3	vehicle	8	0.3581	0.1816	0.7286	0.4647	0.6089

CLIPSeg domain-level values are retained in its case-level evaluation JSON; the main table reports its overall result.

S5. Full A0–A3 Fusion Audit

TABLE S8
A0–A3 by Domain

Domain	n	Variant	F1@0.25	F1@0.50	F1@0.75	Sem. F1@0.25	Obj. IoU	Overseg.
container	72	A0 independent max	0.2357	0.1610	0.0422	0.1500	0.5288	0.3491
container	72	A1 consensus	0.2766	0.1795	0.0608	0.1671	0.5369	0.4417
container	72	A2 hierarchy	0.3096	0.1394	0.0463	0.1670	0.6220	1.1868
container	72	A3 full fusion	0.3237	0.1271	0.0421	0.1575	0.6353	1.3097
daily_object	40	A0 independent max	0.1328	0.0575	0.0392	0.0665	0.2637	0.3550
daily_object	40	A1 consensus	0.1519	0.0575	0.0392	0.0825	0.2706	0.4288
daily_object	40	A2 hierarchy	0.2269	0.0871	0.0273	0.0572	0.3976	1.6058
daily_object	40	A3 full fusion	0.2737	0.0843	0.0398	0.0787	0.4563	1.7224
device	59	A0 independent max	0.1324	0.0695	0.0105	0.0965	0.2300	0.3535
device	59	A1 consensus	0.1828	0.1237	0.0380	0.1354	0.2840	0.4957
device	59	A2 hierarchy	0.3159	0.1842	0.0694	0.1705	0.4947	1.7582
device	59	A3 full fusion	0.3392	0.1953	0.0966	0.1674	0.5735	2.1404

Domain	n	Variant	F1@0.25	F1@0.50	F1@0.75	Sem. F1@0.25	Obj. IoU	Overseg.
furniture	16	A0 independent max	0.0634	0.0000	0.0000	0.0270	0.0453	0.6201
furniture	16	A1 consensus	0.1958	0.0799	0.0114	0.0986	0.1360	0.7655
furniture	16	A2 hierarchy	0.2368	0.1432	0.0514	0.0944	0.2543	1.3274
furniture	16	A3 full fusion	0.2807	0.1408	0.0519	0.0810	0.3343	1.4534
tool_prop	31	A0 independent max	0.0981	0.0441	0.0161	0.0336	0.1314	0.4073
tool_prop	31	A1 consensus	0.1552	0.0931	0.0658	0.0719	0.1899	0.4355
tool_prop	31	A2 hierarchy	0.2455	0.1687	0.0863	0.1240	0.3074	0.9341
tool_prop	31	A3 full fusion	0.2963	0.2047	0.0949	0.1319	0.3278	0.9745
vehicle	8	A0 independent max	0.0000	0.0000	0.0000	0.0000	0.0082	0.1028
vehicle	8	A1 consensus	0.0932	0.0615	0.0192	0.0615	0.0749	0.1691
vehicle	8	A2 hierarchy	0.3207	0.1309	0.0415	0.1759	0.3892	0.5833
vehicle	8	A3 full fusion	0.3581	0.1816	0.0606	0.2255	0.4647	0.6089

All six domain means improve from A0 to A3 at F1@0.25; vehicle n=8 and furniture n=16 remain descriptive.

TABLE S9
Selected Paired Effects

Comparison	Metric	Mean Δ	95% CI	n
A3 full fusion minus A0 independent max	object_iou	0.2213	[0.1891, 0.2556]	226
A3 full fusion minus A0 independent max	part_recall_at_025	0.2555	[0.2248, 0.2874]	226
A3 full fusion minus A0 independent max	part_f1_at_025	0.1622	[0.1325, 0.1913]	226
A3 full fusion minus A0 independent max	part_f1_at_050	0.0652	[0.0337, 0.0965]	226
A3 full fusion minus A0 independent max	part_f1_at_075	0.0392	[0.0186, 0.0590]	226
A3 full fusion minus A0 independent max	mean_matched_boundary_f1_at_050	0.2119	[0.1567, 0.2689]	226
A3 full fusion minus A0 independent max	semantic_f1_at_025	0.0484	[0.0253, 0.0716]	226
A3 full fusion minus A0 independent max	oversegmentation_ratio	1.1693	[0.9944, 1.3683]	226
A3 full fusion minus A2 consensus hierarchy	object_iou	0.0464	[0.0325, 0.0611]	226
A3 full fusion minus A2 consensus hierarchy	part_recall_at_025	0.0528	[0.0367, 0.0705]	226
A3 full fusion minus A2 consensus hierarchy	part_f1_at_025	0.0302	[0.0164, 0.0455]	226
A3 full fusion minus A2 consensus hierarchy	part_f1_at_050	0.0051	[-0.0098, 0.0196]	226

Comparison	Metric	Mean Δ	95% CI	n
A3 full fusion minus A2 consensus hierarchy	part_f1_at_075	0.0099	[-0.0013, 0.0210]	226
A3 full fusion minus A2 consensus hierarchy	mean_matched_boundary_f1_at_050	0.0284	[0.0023, 0.0556]	226
A3 full fusion minus A2 consensus hierarchy	semantic_f1_at_025	0.0019	[-0.0076, 0.0118]	226
A3 full fusion minus A2 consensus hierarchy	oversegmentation_ratio	0.1750	[0.1207, 0.2345]	226

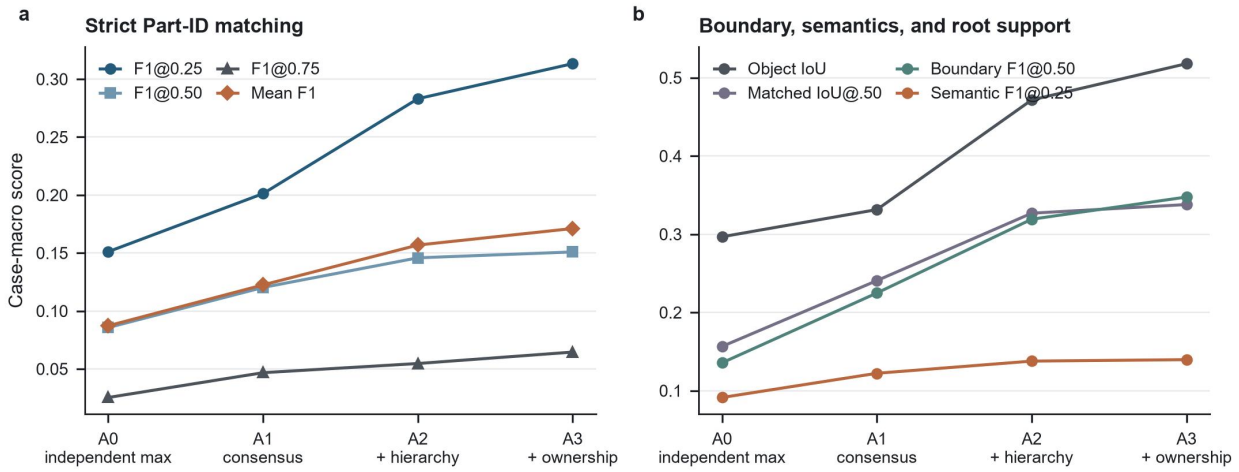


Fig. S2. Earlier locked, frozen-candidate A0–A3 fine Part-ID ablation. A3 improves coarse discovery and root support, whereas gains over A2 at stricter overlap and semantic matching are small.

S6. v0.3.1 Qualitative Audit

The complete plates below retain the PACO reference parts and both public identity layers. They remain in the Supplement as diagnostic audits rather than main-text success illustrations: a case can obtain a moderate overlap score while carrying the wrong routed category or implausible physical groups. The lower-quartile plate is shown separately, and aggregate metrics remain the performance evidence.

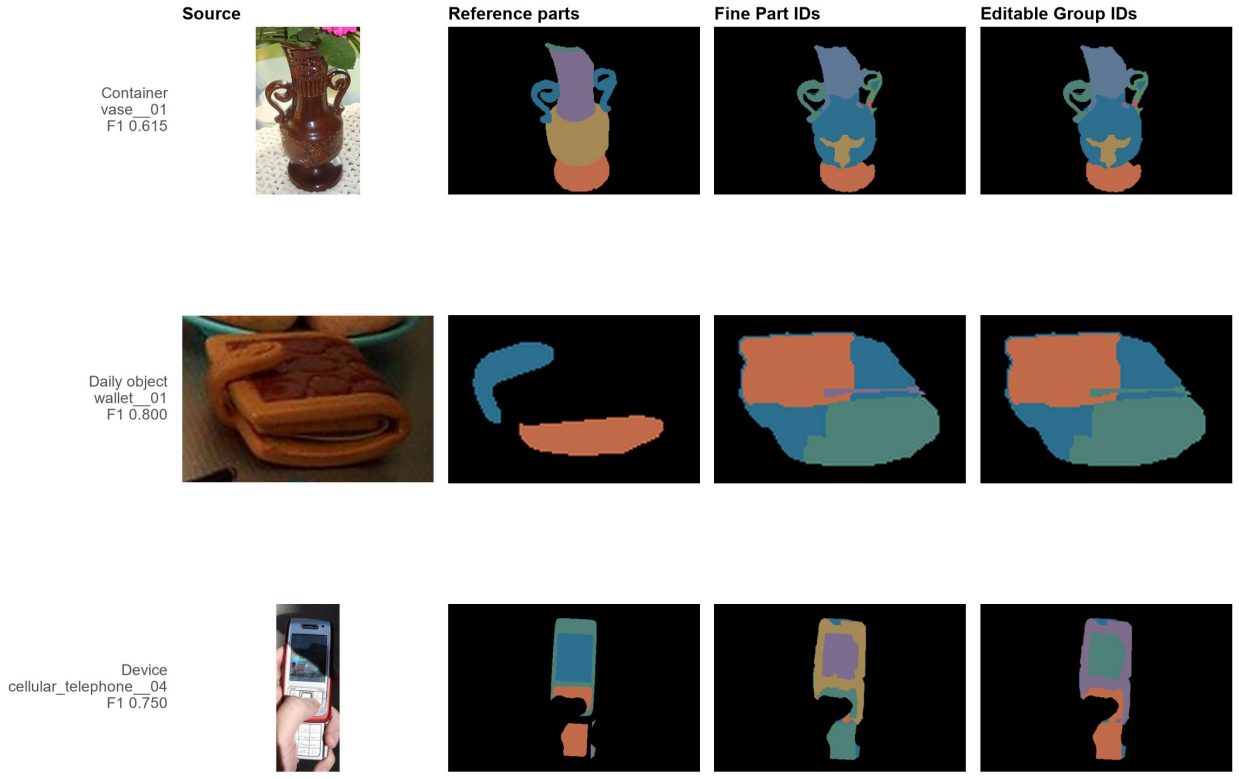


Fig. S3. Upper-quartile readability selection: containers, daily objects, and devices.



Fig. S4. Upper-quartile readability selection: furniture, tools/props, and vehicles.

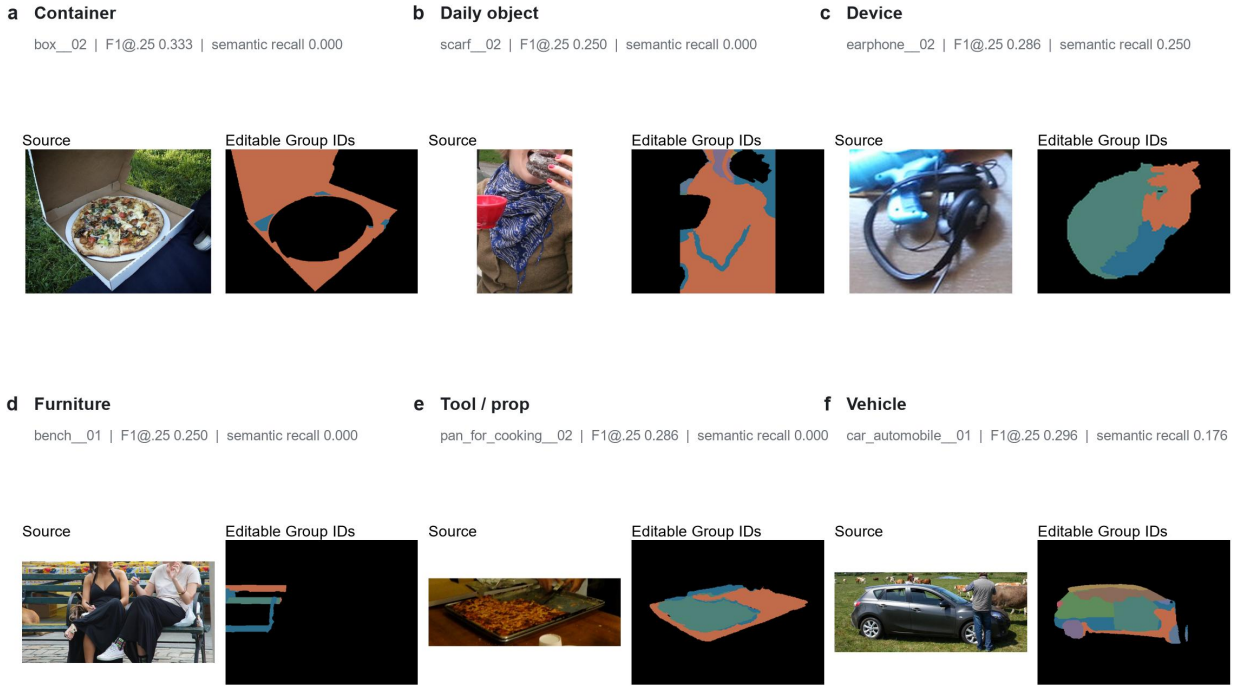


Fig. S5. Cases nearest each domain's lower-quartile editable-group F1@0.25. Failures remain visible and are not used as proof of release quality.

S7. Retrospective Group-Gate and Fine-Fusion Audits

C1-C3 are development-only candidate-retention readouts from stored Group-stage verification diagnostics. They are not the executed input filters of fine-fusion A0-A3. The archived fine-fusion evaluator loads the complete exported candidate pool, including candidates rejected as Group seeds. Thus the original gate-to-final correlations show association between different diagnostic layers, not a causal gate ablation. Section S19.6 now supplies actual Group-gate removals from fixed pre-gate inputs, separately from the fine-fusion sensitivity study.

TABLE S10
Three-Stage Candidate Gate

Condition	n	Candidates	P@0.50	R@0.50	F1@0.50	Zero
C1_semantic	65	5.0154	0.2245	0.1904	0.1749	9
C2_semantic_structure	65	3.0462	0.2621	0.1740	0.1847	14
C3_full_serial_gate	65	2.8769	0.2785	0.1740	0.1874	14

TABLE S11
C3-to-Final Association

C3 metric	Final metric	Spearman ρ	p	n
candidate_f1_at_025	final_part_f1_at_025	0.4734	6.81e-05	65
candidate_f1_at_050	final_part_f1_at_050	0.5443	2.78e-06	65
candidate_recall_at_025	final_part_recall_at_025	0.4310	0.000338	65
candidate_precision_at_050	final_part_precision_at_050	0.5446	2.74e-06	65
candidate_count	final_oversegmentation_ratio	0.4111	0.000669	65

Unadjusted descriptive correlations; they are not additional causal ablations.

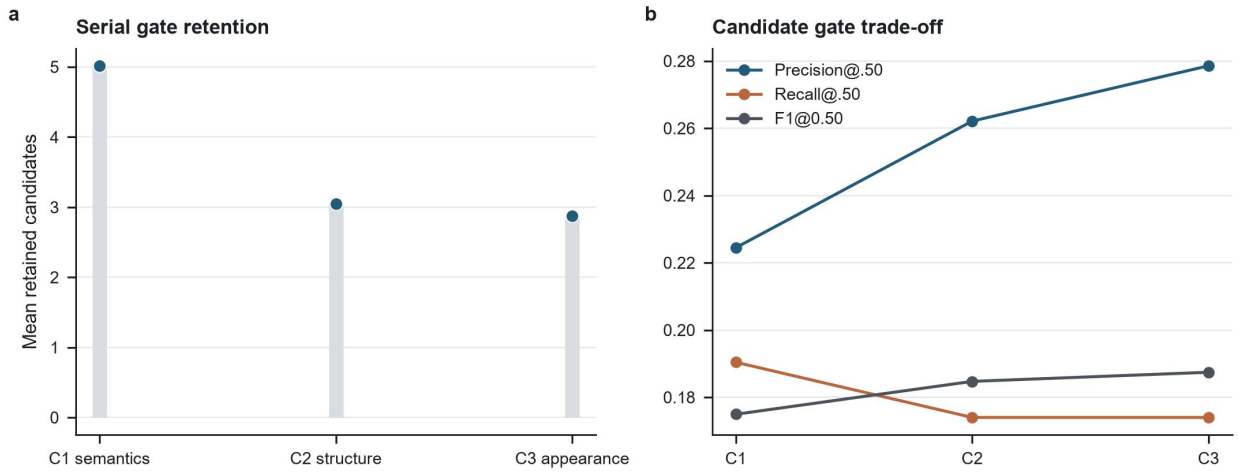


Fig. S6. Development-only linkage between candidate gates and final outputs.

S8. Quality Exit, Coverage, and Failure Taxonomy

TABLE S12
Coverage-Accuracy Trade-off

Policy	Accepted	Coverage	Failure	F1@0.25	F1@0.50	Sem. F1@0.25
No Review Trigger only	61	27.0%	62.3%	0.3810	0.2118	0.2265
Revised rule: No Review Trigger only	61	27.0%	62.3%	0.3810	0.2118	0.2265
No trigger + review	177	78.3%	76.8%	0.3370	0.1530	0.1617
All outputs	226	100.0%	80.5%	0.3133	0.1509	0.1396

TABLE S13
Audited Error-Type Impact

Error type	Affected	Review capture	Δ F1@0.50	Δ Sem. F1@0.25	Δ B-F1@0.50
routing error	37	94.6%	-0.0624	-0.1669	-0.1546
profile error	59	94.9%	-0.0857	-0.1619	-0.2236
weak root	109	80.7%	-0.1027	-0.1449	-0.1759
part discovery failure	144	79.2%	-0.3435	-0.1183	-0.6159
semantic assignment failure	144	80.6%	-0.1024	-0.3106	-0.2264

Twenty-eight No-Review-Trigger cases still fail the semantic threshold. This label denotes only the absence of a mandatory review flag, not verified correctness.

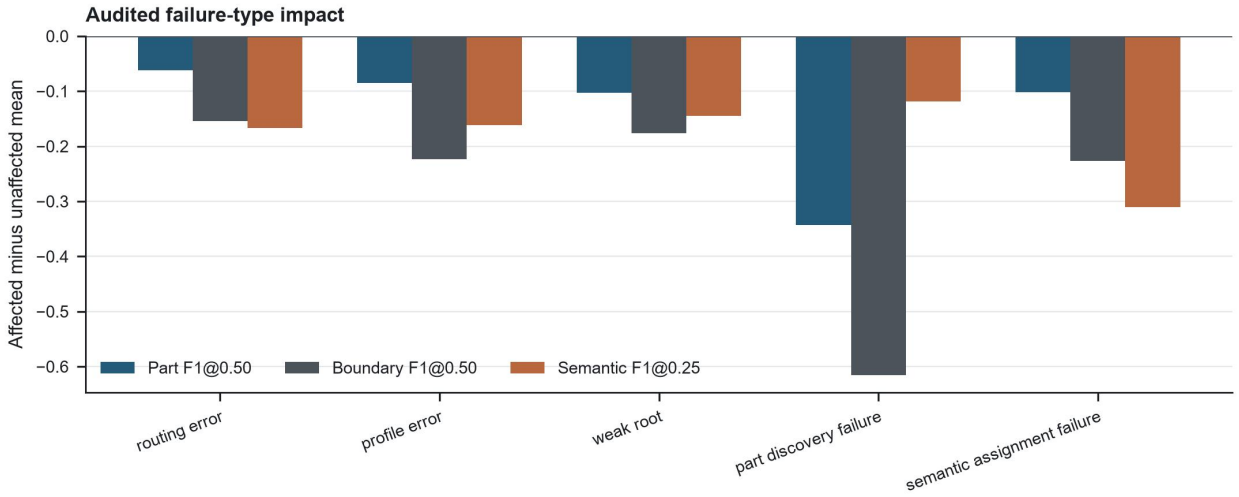


Fig. S7. Metric degradation associated with audited failure types.

S9. Full Sensitivity and Parameter Registry

TABLE S14
All Prespecified One-at-a-Time Perturbations

Parameter	Factor	Base	Tested	F1@0.25	Sem. F1@0.25	Obj. IoU	Fusion s
full_agreement_overlap	0.8	0.55	0.44	0.3133	0.1396	0.5180	0.1695
full_agreement_overlap	0.9	0.55	0.495	0.3133	0.1396	0.5180	0.1713
full_agreement_overlap	1.0	0.55	0.55	0.3133	0.1396	0.5180	0.1719
full_agreement_overlap	1.1	0.55	0.605	0.3133	0.1396	0.5180	0.1712
full_agreement_overlap	1.2	0.55	0.66	0.3133	0.1396	0.5180	0.1732
specificity_minimum_containment	0.8	0.8	0.64	0.3133	0.1396	0.5180	0.1716
specificity_minimum_containment	0.9	0.8	0.72	0.3133	0.1396	0.5180	0.1710
specificity_minimum_containment	1.0	0.8	0.8	0.3133	0.1396	0.5180	0.1727

Parameter	Factor	Base	Tested	F1@0.25	Sem. F1@0.25	Obj. IoU	Fusion s
specificity_minimum_containment	1.1	0.8	0.88	0.3133	0.1396	0.5180	0.1722
specificity_minimum_containment	1.2	0.8	0.96	0.3133	0.1396	0.5180	0.1742
specificity_root_minimum_candidate_score	0.8	0.36	0.288	0.3212	0.1428	0.5293	0.1732
specificity_root_minimum_candidate_score	0.9	0.36	0.324	0.3186	0.1414	0.5239	0.1734
specificity_root_minimum_candidate_score	1.0	0.36	0.36	0.3133	0.1396	0.5180	0.1683
specificity_root_minimum_candidate_score	1.1	0.36	0.396	0.3089	0.1381	0.5112	0.1721
specificity_root_minimum_candidate_score	1.2	0.36	0.432	0.3059	0.1349	0.5024	0.1699
specificity_host_suppression	0.8	0.28	0.224	0.3133	0.1396	0.5180	0.1746
specificity_host_suppression	0.9	0.28	0.252	0.3133	0.1396	0.5180	0.1725
specificity_host_suppression	1.0	0.28	0.28	0.3133	0.1396	0.5180	0.1679
specificity_host_suppression	1.1	0.28	0.308	0.3133	0.1396	0.5180	0.1679
specificity_host_suppression	1.2	0.28	0.336	0.3133	0.1396	0.5180	0.1696
remainder_merge_distance_ratio	0.8	0.025	0.02	0.3133	0.1396	0.5180	0.1668
remainder_merge_distance_ratio	0.9	0.025	0.0225	0.3133	0.1396	0.5180	0.1689
remainder_merge_distance_ratio	1.0	0.025	0.025	0.3133	0.1396	0.5180	0.1669
remainder_merge_distance_ratio	1.1	0.025	0.0275	0.3133	0.1396	0.5180	0.1656
remainder_merge_distance_ratio	1.2	0.025	0.03	0.3133	0.1396	0.5180	0.1654

TABLE S15
Machine-Readable Parameter Registry

Registry	Rows	SHA-256
Global settings	460	68db7a7849553f24016f728c953418acc975500b11c8075ef53bdcc3ffa5aa28
Inventory/profile rules	1515	cbe4e71e3836b19ac8b75737d2f328f946507ced30177a752d4d95194d33d101
Prompt bank	frozen	0c90c30779b670bf17a681df74e5765e751ddb425f9d11a04b7fa2d9da2cddcf

The complete global registry (460 rows) and inventory registry (1,515 rows) are provided in Supplementary Data S1–S2.

S10. Runtime and Reproducibility

TABLE S16
Recorded Hardware

Item	Recorded value
CPU	Intel64 Family 6 Model 154 Stepping 3, GenuineIntel
CPU cores	14 physical / 20 logical
System RAM	39.69 GiB
GPU	NVIDIA GeForce RTX 3060 Laptop GPU
GPU VRAM	6.00 GiB
CUDA runtime reported by PyTorch	12.8

TABLE S17
Recorded Software

Item	Recorded value
operating system	Windows-10-10.0.19045-SP0
python	3.12.2
torch	2.11.0+cu128
torchvision	0.26.0+cu128
transformers	5.12.1
opencv python	5.0.0.93
numpy	2.5.1
git commit	c0f15f72842d7b9239f39c7dcac2f08c95d4923f

TABLE S17A
Model Checkpoints and Execution Settings

Item	Recorded value
Grounding model	IDEA-Research/grounding-dino-tiny
Segmentation model	facebook/sam2.1-hiera-tiny
Dense semantic model	CIDAS/clipseg-rd64-refined
Asset router	google/siglip2-base-patch16-224 compatible local snapshot
Device	cuda
Precision mode	model defaults; no explicit FP16/BF16 autocast override
SAM2 batch size	16
Visual points / crop	18
Visual crop layers	1
Warmup	none
Model loading	included in each command-level total; the benchmark runner launches one isolated process per case
Grounding and SAM2 included	True

No explicit FP16 or BF16 autocast override was applied. Each benchmark case ran in an isolated process, and command-level totals include model loading.

TABLE S17B
Input Object-Crop Resolution (226 Cases)

Measure	Mean	Median	P95	Min-Max
Width (px)	255.8	197.5	589.5	80-640
Height (px)	230.5	195.5	449.5	74-640
Pixel area	66868.2	37660.0	219527.0	6794-296160

These timings concern annotated object crops. Added full-image runtimes are reported in Section S19.12.

TABLE S18
Reproducibility Anchors

Artifact	Value
Algorithm release tag	v0.3.1
Release commit	be543003632fa739d0b63e9acbffb8d91d99b84a
Post-freeze qualitative tag	v0.3.2
Post-freeze qualitative commit	7bb96304706e6ff3e34830c120279f85d7a09017
Test manifest SHA-256	633076503641e81bbc49e796146ff49de637212e9eb1bdc253b547f204f3f548
Benchmark SHA-256	c7a9e99241b80359f40f6f309dd1aae93252e879c794c810c2a11da7e541e464
v0.3.1 regroup summary SHA-256	56c63a0ed82b63b5532fbc7a43051d03f294ce1f6d98f12d2bee3ead7998f2ff
v0.3.1 226-case regroup audit	1,356/1,356 byte-identical core files (226 x 6); 0 mismatch
Locally frozen holdout protocol-record SHA-256	3d9d0191646fc2724f47976513a1c4f6a229c9f87866e3edc09527efd7205d33
Original-holdout summary SHA-256	20063a3a1a6c09e1e0ef84e8d6fd48ed0175a1b8c46c4094b720cbe597014b49
Strict-threshold reporting SHA-256	f4b270dc8aead7603c44a76649eef2a8aa6d10db355926c4d568defc046dd399
Materialization audit SHA-256	a09d282f02a9328497b1d34e45fe2ea49d909bc9fe20c46d292049422c4dd0ec
Evidence-timeline SHA-256	26a04c38becb2cab0803036abf1a5b59b1cf7e01cd38b604b4a51b8129a452bc
Archival evidence commit	c45f9ce3abaecb43e814e86e3a3bb46173fc08a7
Original-holdout determinism	252/252 byte-identical core files (42 x 6); 0 mismatch
Paper fact inventory SHA-256	6523e1b66e9ef578a639506f37acdfb1779f52dc4940cb9f9ddc2c565643609a
Repository visibility	public at the final manuscript audit

TABLE S18A
Holdout Reproduction Command Sequence

Step	Command	Purpose
1	python repro_scripts/prepare_untouched_group_holdout_v031.py	freeze slots and materialize integrity-checked inputs
2	python repro_scripts/run_untouched_group_holdout_v031.py	single-pass inference, scoring, and process-isolated regroup audit
3	python repro_scripts/derive_untouched_holdout_reporting_audit.py	derive strict-threshold, fallback, resize, and timeline records without rerunning predictions

Step	Command	Purpose
4	python repro_scripts/integrate_untouched_holdout_facts.py	merge frozen evidence into the manuscript fact inventory

Run from the delivered evidence root with the recorded Python environment and licensed PACO/LVIS inputs. Per-case commands and source-integrity receipts are preserved in 02_blind_inference_manifest.json and holdout_materialization_audit.json.

S11. Auxiliary Character/Gaussian Compatibility Evidence

The earlier ten-character experiments are retained only to show that persistent IDs can route programmatic Gaussian proxies and two-dimensional known-transform alignment. They are internal, same-asset, downstream compatibility checks. They are not cross-category segmentation evidence, learned depth prediction, photorealistic 3DGS reconstruction, multiview generalization, or physical contact evaluation.

TABLE S19
Bounded Auxiliary Evidence

Experiment	Units	Result	Interpretation
Position recovery	182 parts	crop overlap 0.8646	source-coordinate bookkeeping
Programmatic per-ID Gaussian	153 parts	projection IoU 0.6151	2D orthographic proxy
Exact ID edit selector	153 parts	selection F1 1.0000	ownership contract; not perceptual quality
Rule geometry-scale mapping	153 parts	monotonic fraction 1.0000	rule output; no physical-depth truth
PAGA known 2D transform	200 cases	clean IoU 0.8789	same-mask synthetic recovery
Density 720	153 parts	projection IoU 0.6151	proxy density ablation

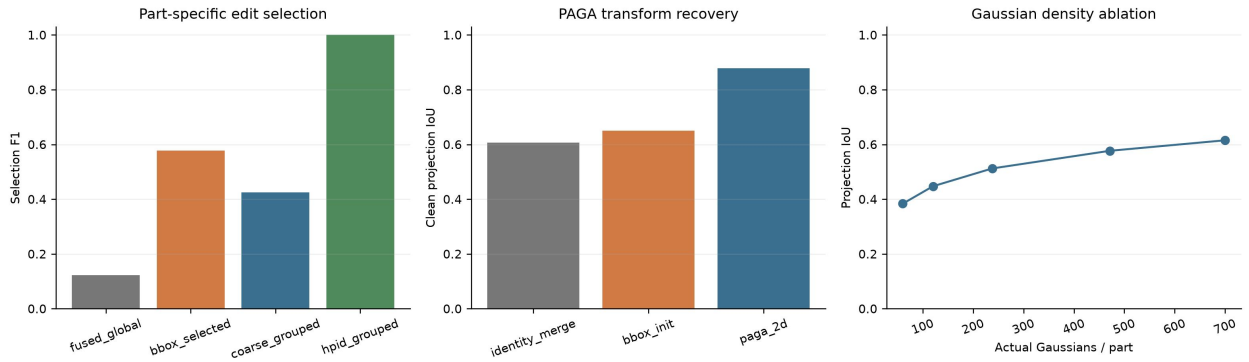


Fig. S8. Auxiliary Gaussian proxy results; these do not enter the cross-category segmentation claim.

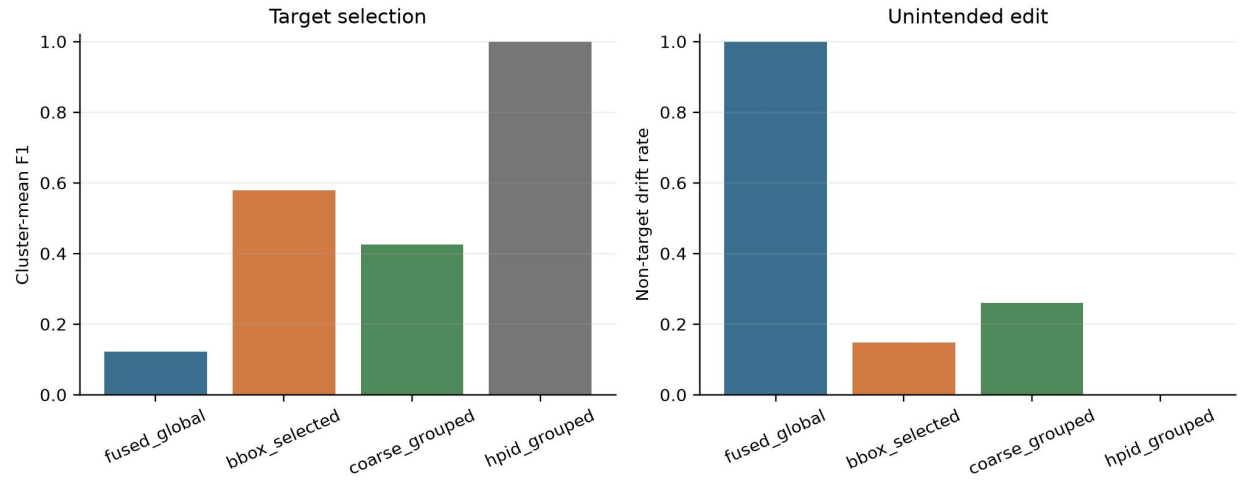


Fig. S9. Exact ID ownership isolates programmatic edit selection; F1=1 is guaranteed by the selector contract.

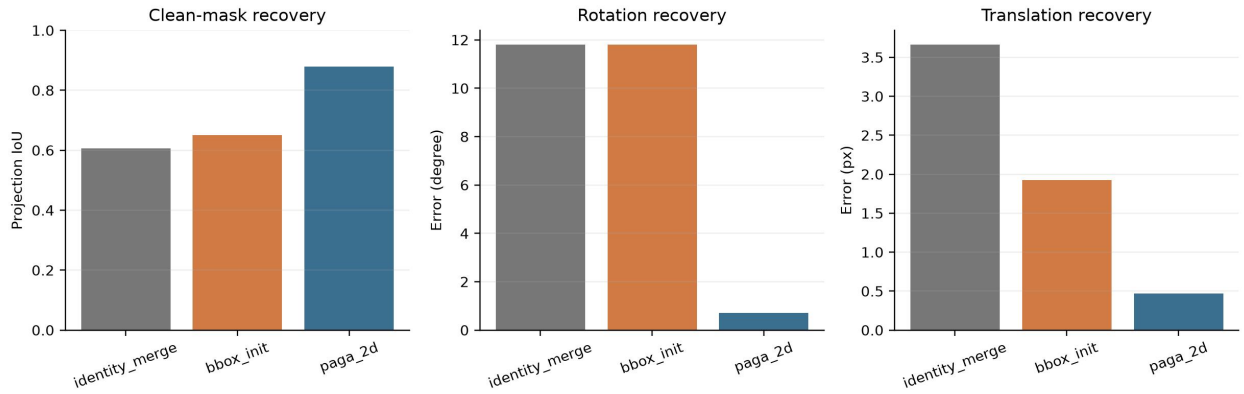


Fig. S10. PAGA recovers synthetic known two-dimensional transforms on the same target masks.

S12. External Paired Effects and Identity Audit

TABLE S20
HPID-Split Minus CLIPSeg: Case-Paired Bootstrap

Metric	Mean difference	95% CI	Includes zero	n
part_f1_at_025	0.0074	[-0.0263, 0.0414]	Yes	226
part_f1_at_050	0.0029	[-0.0299, 0.0345]	Yes	226
part_f1_at_075	0.0057	[-0.0171, 0.0281]	Yes	226
part_f1_mean_025_075	0.0077	[-0.0176, 0.0326]	Yes	226
mean_matched_boundary_f1_at_050	0.0381	[-0.0234, 0.0999]	Yes	226
semantic_f1_at_025	-0.0958	[-0.1283, -0.0644]	No	226
object_iou	-0.1159	[-0.1645, -0.0677]	No	226

Positive values favor HPID-Split. Percentile intervals use 20,000 draws with seed 20260822 and resample the same 226 object-case indices for both methods.

TABLE S21
Representation-Level Audit

Method	Overlap excess	Unassigned root	Exclusive	Persistent IDs	Versioned package
SAM2 raw	0.1025	0.4387	27.9%	Not emitted	Not emitted
SAM2 + NMS	0.0039	0.5151	61.9%	Not emitted	Not emitted
SAM2 + max ownership	0.0000	0.5151	100.0%	Not emitted	Not emitted
Grounded-SAM2	0.0000	0.6346	100.0%	Not emitted	Not emitted
CLIPSeg object-part	1.4272	0.3661	11.9%	Not emitted	Not emitted
HPID-Split A3	0.0000	0.2497	100.0%	Native	Native

Not emitted records the tested implementation's export behavior, not a segmentation error or intrinsic algorithm limitation.

TABLE S22
HPID Package Invariants

Audit item	Result
Valid algorithm-authored export payloads	226/226
Duplicate Part IDs	0
Duplicate Group IDs	0
Invalid assembly-parent references	0
Unresolved part-to-group references	0

Audit item	Result
Canonical rerun exact-map rate	100.0%
Candidate-order exact-map trials	678/678
Post-hoc evaluator sidecars	226/226

paco_evaluation.json was added after inference and is deliberately outside the algorithm-authored manifest; no other export-payload error was observed.

S12.1 Post-Freeze v0.3.8 Engineering Audit

Version 0.3.8 was run after the v0.3.1 evidence freeze to inspect the physical-ownership, bounded boundary-refinement, and illumination-region rejection implementation used for Fig. 6, including paired road-vehicle wheel recovery. Its aggregate and targeted values are retained here for auditability, not as updated primary estimates or evidence that the later implementation outperforms v0.3.1.

TABLE S22A
v0.3.8 Post-Freeze Execution Record

Audit item	Value	Scope
Editable-group F1@0.25	0.3773	Post-freeze v0.3.8 regroup over the engineering packages
Mean matched IoU	0.46415 -> 0.46441	Before/after the bounded engineering changes
Mean Boundary F1@0.25	0.49015 -> 0.49047	Before/after the bounded engineering changes
Wrapped pump-bottle label union IoU	0.8363	Targeted post-inference boundary check
Road-vehicle wheel union IoU	0.5936	Targeted post-inference paired-instance check
Validated packages	226/226	Package/schema validation
Independent full-regroup determinism	1,356/1,356	Core files byte-identical
Code anchor	9b4fa8f0a0504c7ffd68b80ca10771e257ae75c1	v0.3.8

The v0.3.8 results were obtained from post-freeze engineering regrouping and do not establish an improvement over the release results in Table VII. Figure 6 shows the corresponding qualitative examples.

S13. Same-Candidate Postprocessing

All five methods below consume the same 1,951 accepted masks from the 226 locked cases. No proposal generator is rerun. This audit asks whether a simple deterministic postprocessor can match the mask metrics of HPID-Split, and which identity artifacts each route produces.

TABLE S23
Complete Same-Candidate Postprocessing Summary

Method	F1@0.25	F1@0.50	Sem. F1	B-F1@0.25	Obj. IoU	Overlap	Exclusive	Order stable
Raw proposals	0.3176	0.2045	0.1630	0.5242	0.5833	1.0299	5.8%	11.1%
Greedy NMS	0.3464	0.1793	0.1530	0.4707	0.5142	0.0000	100.0%	100.0%
DBSCAN fusion	0.3550	0.1934	0.1536	0.4803	0.5634	0.0000	100.0%	100.0%
Local pairwise CRF	0.2999	0.1533	0.1330	0.4416	0.5150	0.0000	100.0%	100.0%
HPID-Split A3	0.3133	0.1509	0.1396	0.4529	0.5180	0.0000	100.0%	100.0%

Candidate-order stability is computed from fixed permutations of the stored candidates. Persistent hierarchy and package validity remain HPID-specific outputs.

TABLE S24
Selected Paired Effects Against Simple Postprocessors

Comparison	Metric	Mean difference	95% CI
hpid_split_a3_minus_greedy_nms	part_f1_at_025	-0.0331	[-0.0532, -0.0142]

Comparison	Metric	Mean difference	95% CI
hpid_split_a3_minus_greedy_nms	part_f1_at_050	-0.0284	[-0.0479, -0.0096]
hpid_split_a3_minus_greedy_nms	semantic_f1_at_025	-0.0134	[-0.0267, -0.0003]
hpid_split_a3_minus_greedy_nms	object_iou	0.0038	[-0.0154, 0.0230]
hpid_split_a3_minus_dbscan_fusion	part_f1_at_025	-0.0417	[-0.0595, -0.0243]
hpid_split_a3_minus_dbscan_fusion	part_f1_at_050	-0.0425	[-0.0612, -0.0240]
hpid_split_a3_minus_dbscan_fusion	semantic_f1_at_025	-0.0139	[-0.0270, -0.0009]
hpid_split_a3_minus_dbscan_fusion	object_iou	-0.0455	[-0.0641, -0.0287]
hpid_split_a3_minus_local_pairwise_crf	part_f1_at_025	0.0135	[-0.0050, 0.0314]
hpid_split_a3_minus_local_pairwise_crf	part_f1_at_050	-0.0024	[-0.0208, 0.0163]
hpid_split_a3_minus_local_pairwise_crf	semantic_f1_at_025	0.0066	[-0.0070, 0.0207]
hpid_split_a3_minus_local_pairwise_crf	object_iou	0.0030	[-0.0091, 0.0157]

Differences are HPID-Split A3 minus the named comparator; negative values favor the comparator.

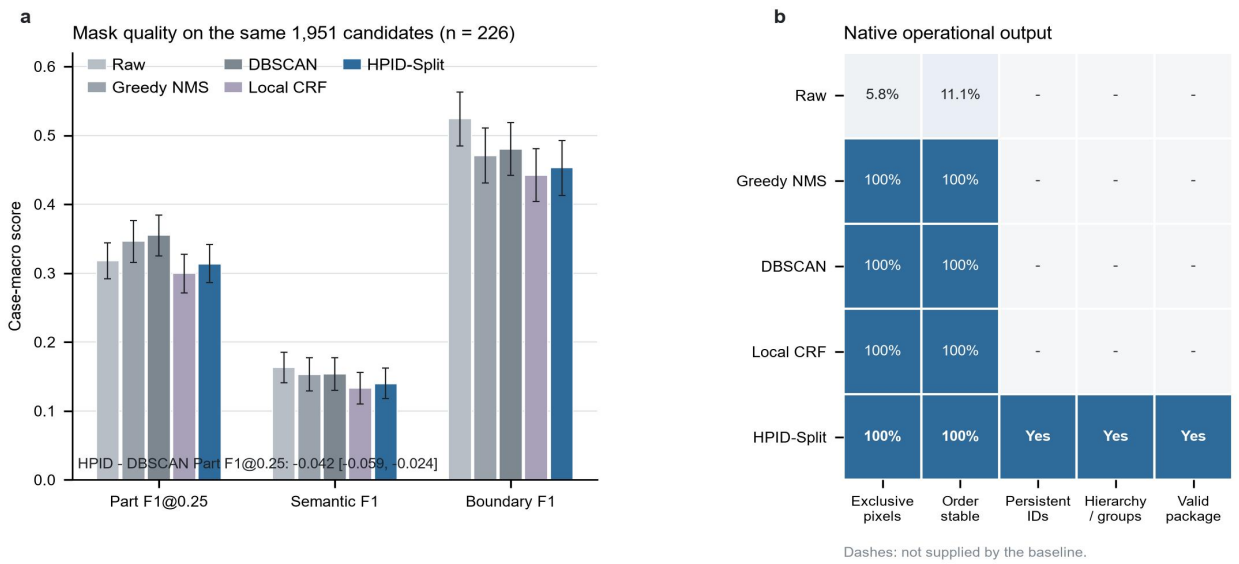


Fig. S11. Mask quality and identity behavior under identical frozen proposals.

S14. Original 37-case Controlled Request-Mask Handoff

The 37-case protocol materializes identity packages from the unoccluded images before constructing defects. The target part name and category are supplied as the task query. Reference-mask geometry defines the defect and scoring target but is withheld from identity selection. With multiple matches, the backend diagnostic selects the maximum by confidence, mask area, then identity; the query still counts as ambiguous. Thus the protocol measures previously computed identity routing and bounded request support, not recognition of a newly occluded part or amodal-support estimation. All original results below are retained; new routing intervals and case transitions are in Section S19. The routing and request-mask expansion is reported separately in S19.13; no additional LaMa outputs were generated.

TABLE S25. Original 37-case Semantic Routing Audit

Representation	Correct unique@0.25	Correct unique@0.50	Wrong unique@0.25	Wrong unique@0.50	Correct in review@0.25	Correct in review@0.50	Unresolved	Ambiguous n
Raw proposals	0.2432	0.0541	0.1351	0.3243	0.6923	0.5385	0.2703	13

Representation	Correct unique@0.25	Correct unique@0.50	Wrong unique@0.25	Wrong unique@0.50	Correct in review@0.25	Correct in review@0.50	Unresolved	Ambiguous n
Greedy NMS	0.2432	0.1351	0.1081	0.2162	0.2500	0.1250	0.4324	8
DBSCAN fusion	0.3784	0.1622	0.1351	0.3514	0.1429	0.0000	0.2973	7
Local pairwise CRF	0.1622	0.0811	0.0811	0.1622	0.5000	0.2143	0.3784	14
HPID Group IDs	0.3784	0.1892	0.1622	0.3514	0.1667	0.0000	0.2973	6

Reference-mask geometry defines the controlled defect and the evaluation target. It is withheld from the identity resolver, which receives the target name and category and searches the pre-defect package. Correctness then requires the selected support to overlap the target at the stated threshold; a name match alone is insufficient.

TABLE S26. Original 37-case Support and Fixed-Backend Diagnostics

Representation	Resolved	Unique	Options	Selected IoU	Req. precision	Req. recall	PSNR	Lock
Raw proposals	0.7297	0.3784	3.6757	0.2313	0.5135	0.4081	17.36	100.0%
Greedy NMS	0.5676	0.3514	3.7838	0.2027	0.4324	0.2995	17.05	100.0%
DBSCAN fusion	0.7027	0.5135	2.8108	0.2334	0.4595	0.3058	16.92	100.0%
Local pairwise CRF	0.6216	0.2432	4.3243	0.2176	0.3784	0.2877	16.84	100.0%
HPID Group IDs	0.7027	0.5405	2.0541	0.2476	0.4865	0.3226	16.62	100.0%

All 37 requests enter each mean. Request precision scores empty U as zero and nonempty U as one because U is a subset of O; it therefore equals the nonempty-request fraction, not semantic precision. PSNR covers O; option counts (S19.10) do not measure human time.

TABLE S27. Original 37-case HPID Group IDs Minus Raw Proposals

Metric	Mean difference	95% CI	n
query_unique	0.1622	[0.0270, 0.2973]	37
review_option_count	-1.6216	[-2.5405, -0.7568]	37
selected_part_iou	0.0163	[-0.0223, 0.0575]	37
completion_request_recall	-0.0854	[-0.2007, 0.0225]	37
visible_lock_exact	0.0000	[0.0000, 0.0000]	37
hidden_region_psnr	-0.7431	[-1.6322, 0.0153]	37

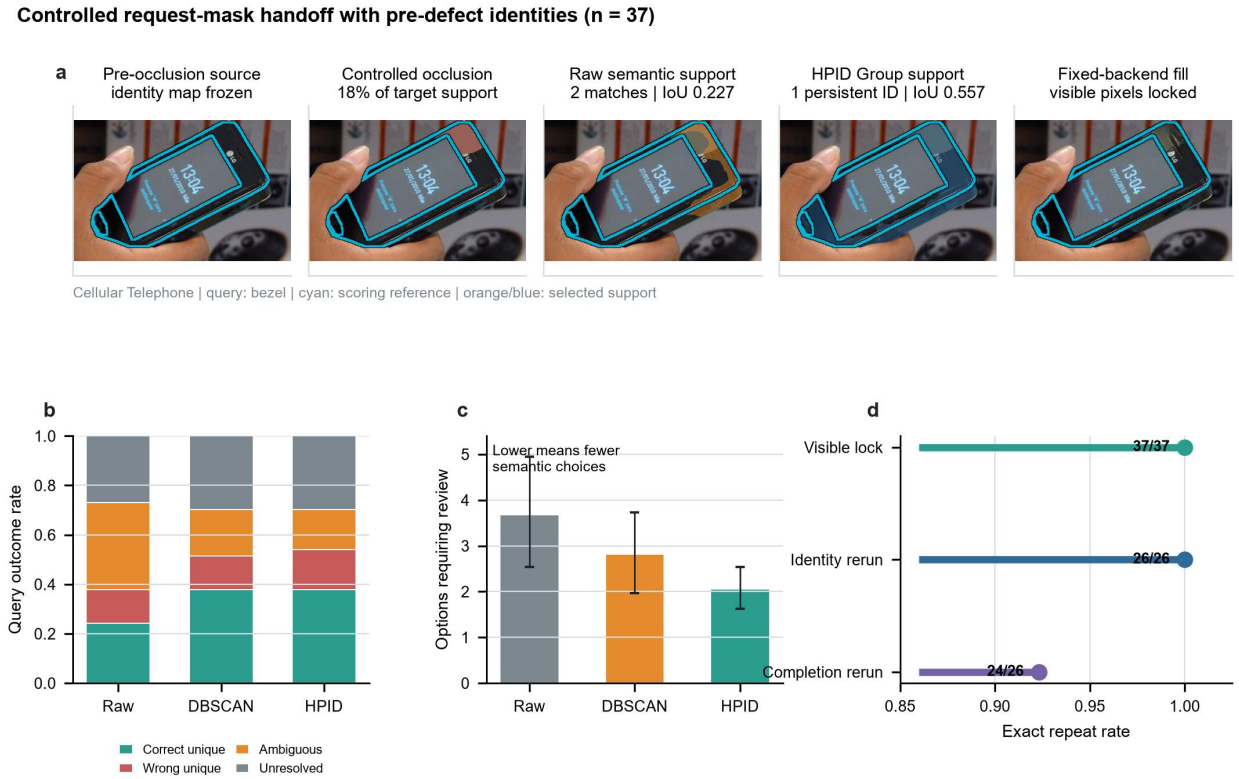


Fig. S12. Original 37-case identity-to-request demonstration, routing outcomes, review options, visible locking and isolated reruns. The rule-selected example and all image diagnostics remain historical; the 253-request routing extension is reported in S19.13.

S15. Automatic-Root Robustness

TABLE S28
Fusion Under Automatic and Perturbed Object Roots

Root	Root IoU	F1@0.25	Sem. F1	B-F1@0.25	Coverage IoU	ID retained	Same-ID IoU
Automatic root	0.6707	0.3133	0.1396	0.4529	0.5180	1.0000	1.0000
Eroded root	0.5736	0.2844	0.1343	0.3927	0.4709	0.9298	0.7316
Dilated root	0.6609	0.3052	0.1340	0.4194	0.5058	0.8934	0.7762
Translated root	0.6046	0.2868	0.1340	0.4012	0.4836	0.8919	0.6951
Reference-root diagnostic	1.0000	0.3476	0.1442	0.4811	0.6723	0.8196	0.6006

Reference-root values are fixed-candidate diagnostic contrasts, not guaranteed upper bounds. Every condition keeps the annotation-derived object crop box.

TABLE S29
Performance by Automatic-Root IoU Stratum

Root-IoU bin	n	Mean root IoU	F1@0.25	Sem. F1	Root coverage
[0.00,0.50)	59	0.2328	0.1615	0.0419	0.2326
[0.50,0.70)	41	0.6174	0.3200	0.1374	0.6174
[0.70,0.85)	35	0.7872	0.3405	0.1379	0.7870
[0.85,1.00]	91	0.9339	0.3983	0.2046	0.9335

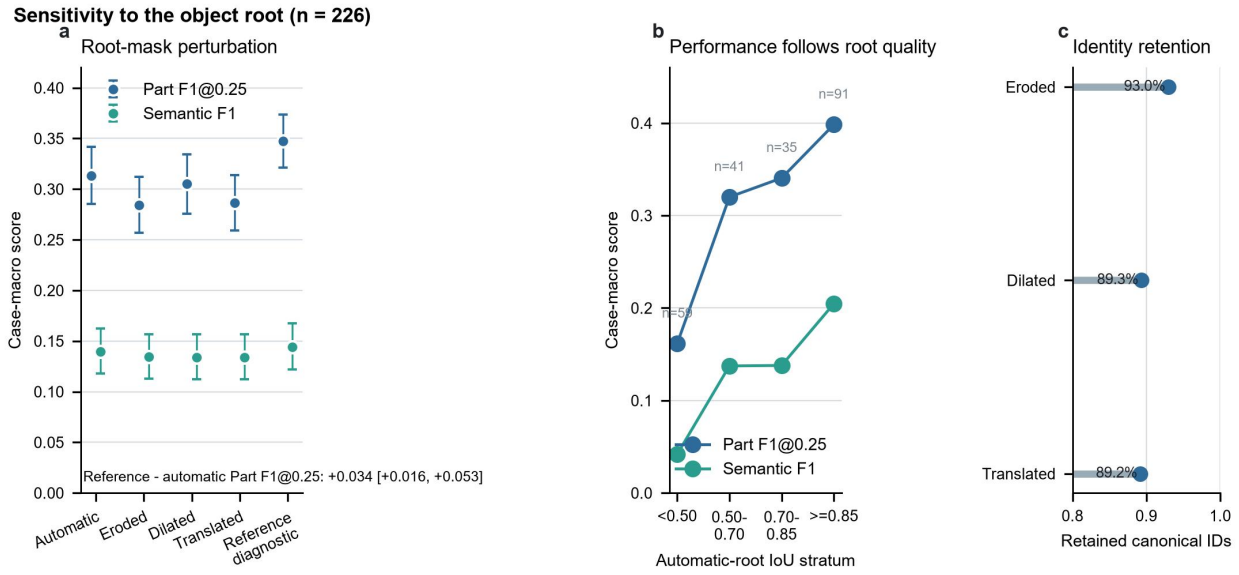


Fig. S13. Part quality and identity retention under automatic-root perturbations.

S16. Executability Audit for Recent Part Methods

TABLE S30
Recorded Executability Audit

Method	Official commit/input	Executed	Execution evidence
PartCATSeg	4a9a88288659f49bc39ea4ee756dd45036552638	Yes: 42/42	Official VOC checkpoint; fixed PACO456 vocabulary; all 42 original crops completed. Cross-dataset adapted comparison, S19.14/Table S53.
HOPS	6273c65324c1162e8b6954edae3303b696590231	No	Earlier entrypoint probe failed at the Detectron2 import. No subsequent verified trained accuracy run; the local failure does not imply global unavailability.
KPS	Not supplied	No	No verified local trained execution. Implementation availability is not inferred from this local audit.

The earlier PartCATSeg and HOPS entrypoint probes stopped at the Detectron2 import before trained inference; their commands and tracebacks remain in the historical audit. We subsequently resolved the PartCATSeg environment and completed trained inference on all 42 original holdout crops, using official source commit 4a9a88288659f49bc39ea4ee756dd45036552638 and its released VOC checkpoint (S19.14). HOPS has no subsequent verified trained run, and KPS remains unexecuted. These are local execution records, not claims of global implementation or checkpoint unavailability. Official sources are https://github.com/kaist-cvml/part-catseg and https://github.com/TJU-IDVLab/HOPS.

S17. Validated and Prospective Uses

TABLE S31
Application Boundary

Use	Evidence status	Required extension
Persistent semantic Part/Group IDs	Validated	226 cases; export and determinism audits
Controlled request-mask handoff	Validated	253 routing requests; 37 LaMa cases; pre-defect identities
Single-image occluded-part restoration	Prospective integration	Requires hidden-support estimator and amodal truth
Pre-3D semantic organization	Prospective integration	No 3D generation evaluated
Out-of-frame or occluded extent	Not validated	Requires amodal/generative model and truth

Use	Evidence status	Required extension
Root/stem/leaf monitoring	Not validated	Requires agricultural data and domain inventory

S18. Data Availability

- Code and earlier evidence are available at https://github.com/pennchenj-lang/ID_Part. The revision evidence archive (10 October 2026; commit 0b294f1b) contains 260 scientific source files plus eight release-metadata files, including per-case tables, frozen protocols, analysis code, audit records and checksum mappings. Its bilingual index maps all 14 review comments and links the earlier Group-gate evidence at docs/evidence/revision_20261008/evidence/gate_ablation. This evidence-release commit does not identify a single algorithm version: frozen source and checkpoint versions are recorded per protocol. Original images, full candidate-mask pools, model weights and the runtime environment are not included. Full pixel-level replay therefore also requires the original inputs and frozen environment under their respective access and licence conditions. Historical releases remain unchanged.

S19. Post-review Diagnostics

Sections S19.1-S19.11 retain earlier frozen-candidate diagnostics on the 226 test and 42 holdout cases. S19.12 reruns the frozen pipeline on all 42 full source images. S19.13 expands routing to 253 image-level requests, one per distinct source image drawn from those existing benchmarks, using archived proposals and common frozen postprocessing. These are retrospective extensions, not new untouched samples or a retuned release. Statistical sample size is the number of independent source images, not the number of methods or settings. Section S19.14 reports the executed official PartCATSeg transfer comparison.

S19.1 Scalar and source-weight sensitivity

The audit comprises 50 settings on 226 cases, or 11,300 case-setting evaluations. Single-parameter sweeps include removal, half-strength and +/-20% changes where applicable; 16 joint corners perturb detail bonus, parent-support base, hierarchy exponent and conflict penalty by +/-20%. The conflict penalty is one minus the host multiplier, so its removal sets that multiplier to one but retains the child evidence floor. Disabling specificity-aware ownership removes both operations. The parent-support base also supplies the direct-detail gate threshold; its sweep is therefore not isolated to parent attenuation. Other runs remove individual components or set source reliability to one, 0.8 times or 1.2 times its stored value; scaled values are clipped to [0,1], and the interventions affect every use of reliability in fine fusion. Candidates remain fixed and Group construction is not rerun. Paired 10,000-draw case-bootstrap intervals (seed 20261008) are pointwise exploratory 95% intervals without adjustment for 49 release contrasts per endpoint. The sample size is 226 cases, not 11,300 independent observations. No new setting replaces the release.

TABLE S32. All frozen-candidate settings on 226 cases

Setting	Changed maps	Part F1@.25	Change vs release [95% CI]
release	0	0.3133	+0.0000 [+0.0000, +0.0000]
detail bonus=0	174	0.3062	-0.0071 [-0.0184, +0.0034]
detail bonus=0.04	162	0.3106	-0.0027 [-0.0126, +0.0064]
detail bonus=0.064	149	0.3127	-0.0006 [-0.0062, +0.0045]
detail bonus=0.096	140	0.3192	+0.0059 [-0.0006, +0.0148]
detail bonus=0.12	149	0.3222	+0.0089 [+0.0009, +0.0188]
orphan support=0	27	0.3116	-0.0017 [-0.0045, +0.0004]
orphan support=0.06	1	0.3133	+0.0000 [+0.0000, +0.0000]
orphan support=0.096	0	0.3133	+0.0000 [+0.0000, +0.0000]
orphan support=0.144	0	0.3133	+0.0000 [+0.0000, +0.0000]
orphan support=0.18	0	0.3133	+0.0000 [+0.0000, +0.0000]
hierarchy strength=0	32	0.3137	+0.0004 [-0.0015, +0.0027]
hierarchy strength=0.325	2	0.3133	+0.0000 [+0.0000, +0.0000]
hierarchy strength=0.52	0	0.3133	+0.0000 [+0.0000, +0.0000]
hierarchy strength=0.78	0	0.3133	+0.0000 [+0.0000, +0.0000]
hierarchy strength=0.975	0	0.3133	+0.0000 [+0.0000, +0.0000]

Setting	Changed maps	Part F1@.25	Change vs release [95% CI]
conflict penalty=0	125	0.3072	-0.0061 [-0.0143, +0.0007]
conflict penalty=0.36	2	0.3133	+0.0000 [+0.0000, +0.0000]
conflict penalty=0.576	0	0.3133	+0.0000 [+0.0000, +0.0000]
conflict penalty=0.864	0	0.3133	+0.0000 [+0.0000, +0.0000]
conflict penalty=0.95	0	0.3133	+0.0000 [+0.0000, +0.0000]
direct gate threshold=0.304	164	0.3131	-0.0002 [-0.0032, +0.0022]
direct gate threshold=0.456	175	0.3049	-0.0084 [-0.0160, -0.0015]
direct gate margin=0.656	156	0.3311	+0.0178 [+0.0056, +0.0320]
direct gate margin=0.984	175	0.3060	-0.0073 [-0.0189, +0.0038]
joint 0.8 0.8 0.8 0.8	149	0.3127	-0.0006 [-0.0062, +0.0045]
joint 0.8 0.8 0.8 1.2	149	0.3127	-0.0006 [-0.0062, +0.0045]
joint 0.8 0.8 1.2 0.8	149	0.3127	-0.0006 [-0.0062, +0.0045]
joint 0.8 0.8 1.2 1.2	149	0.3127	-0.0006 [-0.0062, +0.0045]
joint 0.8 1.2 0.8 0.8	149	0.3127	-0.0006 [-0.0062, +0.0045]
joint 0.8 1.2 0.8 1.2	149	0.3127	-0.0006 [-0.0062, +0.0045]
joint 0.8 1.2 1.2 0.8	149	0.3127	-0.0006 [-0.0062, +0.0045]
joint 0.8 1.2 1.2 1.2	149	0.3127	-0.0006 [-0.0062, +0.0045]
joint 1.2 0.8 0.8 0.8	140	0.3192	+0.0059 [-0.0006, +0.0148]
joint 1.2 0.8 0.8 1.2	140	0.3192	+0.0059 [-0.0006, +0.0148]
joint 1.2 0.8 1.2 0.8	140	0.3192	+0.0059 [-0.0006, +0.0148]
joint 1.2 0.8 1.2 1.2	140	0.3192	+0.0059 [-0.0006, +0.0148]
joint 1.2 1.2 0.8 0.8	140	0.3192	+0.0059 [-0.0006, +0.0148]
joint 1.2 1.2 0.8 1.2	140	0.3192	+0.0059 [-0.0006, +0.0148]
joint 1.2 1.2 1.2 0.8	140	0.3192	+0.0059 [-0.0006, +0.0148]
joint 1.2 1.2 1.2 1.2	140	0.3192	+0.0059 [-0.0006, +0.0148]
No parent support	32	0.3137	+0.0004 [-0.0015, +0.0027]
No direct gate	178	0.3015	-0.0118 [-0.0243, +0.0002]
No specificity ownership	163	0.2831	-0.0302 [-0.0454, -0.0163]
No hierarchical duplicate suppression	0	0.3133	+0.0000 [+0.0000, +0.0000]
No remainder attachment	0	0.3133	+0.0000 [+0.0000, +0.0000]
No consensus	78	0.3019	-0.0114 [-0.0180, -0.0055]
source reliability uniform	188	0.3321	+0.0187 [+0.0049, +0.0340]
source reliability 0.8x	193	0.3097	-0.0036 [-0.0141, +0.0074]
source reliability 1.2x	185	0.3284	+0.0151 [+0.0033, +0.0280]

Relative to release Part F1@0.25 of 0.3133, individual +/-20% changes to the conflict penalty (0.576/0.864), parent-support base (0.096/0.144) or hierarchy exponent (0.52/0.78) left all 226 maps unchanged. Detail bonuses of 0.064 and 0.096 yielded F1 0.3127 and 0.3192 and changed 149 and 140 maps, respectively. All 16 joint corners spanned the same F1 range. Changed-map counts include any pixel-label difference, including renumbering, and do not by themselves measure loss of persistent identity. Local stability of the aggregate score need not imply identical maps.

Removing the detail bonus, host penalty or parent attenuation changed F1 by -0.0071 [-0.0184, 0.0034], -0.0061 [-0.0143, 0.0007] and +0.0004 [-0.0015, 0.0027], respectively. Their intervals include zero. Removing the complete specificity-aware ownership module lowered F1 to 0.2831, a difference of -0.0302 [-0.0454, -0.0163]. This supports the combined mechanism

in this cohort, not the necessity or optimality of each scalar. A detail bonus of 0.12 yielded 0.3222, a -20% direct-gate margin change yielded 0.3311, and uniform reliability yielded 0.3321 (+0.0187 [0.0049, 0.0340]); these outcomes preclude a claim of general insensitivity or an optimal released setting. Removing hierarchical duplicate suppression or remainder attachment left maps unchanged in this frozen pool, which does not establish their behavior on other inputs. Consensus removal reduced F1 by 0.0114. These are retrospective fixed-candidate diagnostics, not evidence excluding development-set overfitting or validating upstream and Group-stage robustness.

TABLE S33. Recorded source families and effective weight ranges

Family	n	Stored multiplier range	Effective strength range
Grounded-SAM2	258	0.6564 to 0.9979	0.3779 to 0.9606
Grounded-SAM2/profile-dense	39	0.2779 to 0.7356	0.1317 to 0.6650
Grounded-SAM2/profile-refine	474	0.6610 to 1.1366	0.4118 to 0.7790
hpid-appearance-contour	42	0.6200 to 0.6800	0.4436 to 0.6077
hpid-repetitive-physical-detail-v1	53	0.7600 to 0.7600	0.5594 to 0.7349
hpid-structural-fusion-v1	2	0.7400 to 0.7400	0.6809 to 0.6976
hpid-structural-fusion-v2	3	0.7400 to 0.7400	0.6854 to 0.6966
hpid-topology-v1	13	0.8200 to 0.8200	0.5097 to 0.6481
SAM2 AMG	1067	0.5687 to 0.8200	0.4517 to 0.6873

All observed agreement factors equal one. These are engineering strengths rather than calibrated posterior probabilities; within-family maximum aggregation prevents exact duplicate accumulation but does not remove errors shared between families.

S19.2 Spatial diagnosis of the DBSCAN gap

DBSCAN and HPID receive the same complete exported masks accepted for fine fusion, before the named Group-stage verification. The shared-input replay reproduces all archived numeric evaluator endpoints and all 226 HPID fine maps. Both predict a mean 5.6504 scored parts. DBSCAN obtains 397 matched references at IoU 0.25 versus 354 for HPID; case-macro recall is 0.4078 versus 0.3656, precision 0.3749 versus 0.3343, and matched IoU 0.4490 versus 0.4124. DBSCAN has higher F1 in 83 cases, HPID in 49, and 94 tie. Uniform reliability improves HPID F1 but remains below DBSCAN. The actual fine-stage trace in Section S19.7 follows all 990 reference parts; it does not blame shared input filters or later Group gates for this fine-ID difference.

A taxonomy-clustered bootstrap samples the 60 categories with replacement and retains all cases of each selected category. Across 10,000 draws, seed 20261008, the HPID-minus-DBSCAN case-macro F1 difference is -0.0417 with 95% interval [-0.0601, -0.0244], agreeing with the original case-paired result. Counts of matched parts are descriptive and are not treated as independent sample units.

S19.3 Original 37-case routing uncertainty and transitions

Each case has one target. Unique-correct, wrong-unique, ambiguous, and unresolved states are mutually exclusive at each IoU threshold. Wilson intervals are pointwise 95% intervals; they are not simultaneous multinomial bounds. Conditional correct-in-review intervals use only ambiguous queries. With 37 cases, the intervals are wide and should not be read as deployment guarantees.

TABLE S34. Original 37-case routing intervals at IoU 0.25 and 0.50

Method	Endpoint	Count	Rate [95% CI]
DBSCAN	unique and correct 025	14/37	0.378 [0.241, 0.539]
DBSCAN	wrong unique 025	5/37	0.135 [0.059, 0.280]
DBSCAN	query ambiguous	7/37	0.189 [0.095, 0.342]
DBSCAN	query unresolved	11/37	0.297 [0.175, 0.458]
DBSCAN	unique and correct 050	6/37	0.162 [0.077, 0.311]
DBSCAN	wrong unique 050	13/37	0.351 [0.218, 0.512]
DBSCAN	correct in review set 025	1/7	0.143 [0.026, 0.513]
DBSCAN	correct in review set 050	0/7	0.000 [0.000, 0.354]

Method	Endpoint	Count	Rate [95% CI]
greedy_nms	unique and correct 025	9/37	0.243 [0.134, 0.401]
greedy_nms	wrong unique 025	4/37	0.108 [0.043, 0.247]
greedy_nms	query ambiguous	8/37	0.216 [0.114, 0.372]
greedy_nms	query unresolved	16/37	0.432 [0.287, 0.591]
greedy_nms	unique and correct 050	5/37	0.135 [0.059, 0.280]
greedy_nms	wrong unique 050	8/37	0.216 [0.114, 0.372]
greedy_nms	correct in review set 025	2/8	0.250 [0.071, 0.591]
greedy_nms	correct in review set 050	1/8	0.125 [0.022, 0.471]
HPID	unique and correct 025	14/37	0.378 [0.241, 0.539]
HPID	wrong unique 025	6/37	0.162 [0.077, 0.311]
HPID	query ambiguous	6/37	0.162 [0.077, 0.311]
HPID	query unresolved	11/37	0.297 [0.175, 0.458]
HPID	unique and correct 050	7/37	0.189 [0.095, 0.342]
HPID	wrong unique 050	13/37	0.351 [0.218, 0.512]
HPID	correct in review set 025	1/6	0.167 [0.030, 0.564]
HPID	correct in review set 050	0/6	0.000 [0.000, 0.390]
local_pairwise_crf	unique and correct 025	6/37	0.162 [0.077, 0.311]
local_pairwise_crf	wrong unique 025	3/37	0.081 [0.028, 0.213]
local_pairwise_crf	query ambiguous	14/37	0.378 [0.241, 0.539]
local_pairwise_crf	query unresolved	14/37	0.378 [0.241, 0.539]
local_pairwise_crf	unique and correct 050	3/37	0.081 [0.028, 0.213]
local_pairwise_crf	wrong unique 050	6/37	0.162 [0.077, 0.311]
local_pairwise_crf	correct in review set 025	7/14	0.500 [0.268, 0.732]
local_pairwise_crf	correct in review set 050	3/14	0.214 [0.076, 0.476]
Raw	unique and correct 025	9/37	0.243 [0.134, 0.401]
Raw	wrong unique 025	5/37	0.135 [0.059, 0.280]
Raw	query ambiguous	13/37	0.351 [0.218, 0.512]
Raw	query unresolved	10/37	0.270 [0.154, 0.430]
Raw	unique and correct 050	2/37	0.054 [0.015, 0.177]
Raw	wrong unique 050	12/37	0.324 [0.196, 0.485]

Method	Endpoint	Count	Rate [95% CI]
Raw	correct in review set 025	9/13	0.692 [0.424, 0.873]
Raw	correct in review set 050	7/13	0.538 [0.291, 0.768]

TABLE S35. Original 37-case HPID domain-stratified routing at IoU 0.25

Domain	Correct unique	Wrong unique	Ambiguous	Unresolved
container	4/10 [0.17, 0.69]	2/10 [0.06, 0.51]	1/10 [0.02, 0.40]	3/10 [0.11, 0.60]
daily_object	0/3 [0.00, 0.56]	1/3 [0.06, 0.79]	0/3 [0.00, 0.56]	2/3 [0.21, 0.94]
device	4/8 [0.22, 0.78]	1/8 [0.02, 0.47]	2/8 [0.07, 0.59]	1/8 [0.02, 0.47]
furniture	1/4 [0.05, 0.70]	0/4 [0.00, 0.49]	0/4 [0.00, 0.49]	3/4 [0.30, 0.95]
tool_prop	3/8 [0.14, 0.69]	1/8 [0.02, 0.47]	2/8 [0.07, 0.59]	2/8 [0.07, 0.59]
vehicle	2/4 [0.15, 0.85]	1/4 [0.05, 0.70]	1/4 [0.05, 0.70]	0/4 [0.00, 0.49]

The 37 targets cover 32 actual categories, with one to three cases per category. These category outcomes are descriptive, not reliable category-level performance estimates. Complete raw, DBSCAN, and HPID category/domain counts and Wilson intervals are provided in routing_stratified.csv. Results for every category, including all observed failures, are reported separately in routing_stratified.csv.

TABLE S36. All six wrong-unique HPID cases in the original 37 requests

Case / target	Target IoU	Target precision / recall	Largest reference overlap	Diagnostic
telephone__u01 telephone / bezel	0.0406	0.0834 / 0.0733	back_cover 0.2273	Insufficient target support
bottle__u01 bottle / base	0.0000	0.0000 / 0.0000	label 0.0176	Insufficient target support
box__u01 box / side	0.0069	0.0087 / 0.0327	lid 0.9302	Different annotated part
car_automobile__u02 car_(automobile) / windshield	0.2171	1.0000 / 0.2171	windshield 0.2171	Insufficient target support
shoe__u01 shoe / vamp	0.2069	0.2673 / 0.4780	quarter 0.2189	Insufficient target support
screwdriver__u01 knife / blade	0.0130	0.0192 / 0.0387	handle 0.9294	Different annotated part

All six original wrong-unique outputs match the normalized query name but fail target IoU 0.25. Box-side support overlaps the annotated lid at IoU 0.9302; knife-blade support overlaps the handle at 0.9294. The other four have insufficient target support. The second car has precision 1.0 and recall 0.2171; bottle root coverage is 0.0048 and its selected Group has no target intersection. Telephone, bottle and knife already lack a correct raw option. The screwdriver-named slot is evaluated as its actual category, knife. These are reference-grounded spatial diagnoses, not independent human semantic adjudications or proof of an alias-normalization error.

The original six failures represent 6/37 requests (16.22%; Wilson 95% CI [7.65%, 31.14%]) and 6/20 unique returns. The expanded cohort has 36/253 wrong-unique outcomes (14.23%; [10.46%, 19.07%]) and 36/69 wrong outcomes among unique returns (52.17%; [40.59%, 63.53%]). These are different denominators. The 25 raw-to-HPID losses of a correct matching option in the expanded analysis form a different subset and are not a classification of the 36 wrong-unique outcomes.

TABLE S37. Fate of the nine raw ambiguous requests containing a correct option in the original cohort

Same request	Raw best IoU	Fine best IoU	Group best IoU	Group state
laptop_computer__u01	0.8678	0.6751	0.6707	unique correct
calculator__u01	0.6421	0.5314	0.5567	unique correct
bench__u01	0.7492	0.7459	0.8896	unique correct
box__u01	0.5415	0.0070	0.0069	unique wrong
bicycle__u01	0.4569	0.4463	0.4464	ambiguous

Same request	Raw best IoU	Fine best IoU	Group best IoU	Group state
car_automobile__u02	0.7048	0.2203	0.2171	unique wrong
wrench__u01	0.9527	0.8859	0.7212	unique correct
knife__u01	0.6921	0.7687	0.7725	unique correct
scissors__u01	0.2546	0.1419	0.1419	ambiguous

Exact replay reproduces all 37 fine maps and Group maps, 111 method/request routing rows, and 111 selected masks. The 23 correct raw candidates across 18 requests all survive replayed input deduplication, caps, and hierarchy-duplicate filtering. Sixteen are nominated to the Group predicate, 15 accepted and one rejected. Five of the original nine ambiguous queries with a correct option become correctly unique, two wrongly unique, and two remain ambiguous, with a correct option retained only for the bicycle. Thus the conditional ratios 9/13 and 1/6 mix successful disambiguation with lost target support.

Pixel-flow tracing follows the same candidates rather than comparing endpoint percentages. For the box correct candidate, 6,437 target-intersection pixels are finally owned by bottom (6,157), side (270), and label (10). The second-car candidate has 659 target pixels, finally assigned to windshield (203), window (348), roof (83), body (7), and background (18). Both pass Group verification, and target IoU is already below threshold at fine-ID output. For scissors, 3,023 target pixels end in handle (1,407), body (1,588), and blade (28); the candidate fails Group verification, but support loss precedes it and another candidate verifies the handle name. For the raw-unique shoe, Group assembly improves target IoU from 0.0934 to 0.2069 but cannot reach the threshold. DBSCAN cluster labels can also move a retained candidate outside the requested semantic set. Full per-candidate retention, cluster membership, and conserved pixel destinations are supplied in request_candidate_trace.csv and request_candidate_pixel_flows.csv.

The observed pixel destinations locate where support is represented after fusion; they do not isolate an individual gate, ownership coefficient, extraction step or cleanup operation. Group construction can rebuild partitions as well as merge Parts. A lost matching option therefore need not mean that its pixels were deleted, and a Group rejection need not explain earlier loss. The original nine correct raw ambiguous queries, all six wrong-unique reference comparisons and the complete source-data traces are retained.

The evaluated Group adapter reads the exported ID, semantic name and indexed mask, assigns confidence 1.0, and does not pass review_required to the resolver. All six original selected Groups have review_required=false. The resolver counts normalized name matches; exactly one enters the unique branch, independently of geometric correctness. Confidence, mask area and identity are used for ranking when multiple matches exist, without changing their ambiguous status. The highest-ranked match supplies the request mask in the controlled handoff even for an ambiguous query. The fixed confidence value is not evidence of calibrated certainty and is not the cause of a singleton choice.

We reproduced all 37 stored HPID routing endpoints and selected masks, then tested ten settings for each original wrong-unique case. These 60 observations include unchanged baselines and controls retaining the original false flag or confidence of one. Flipping the exported review_required flag between false and true left the adapter output and routing unchanged. Setting the singleton confidence to zero or one likewise left the selection unchanged. Empty, full-image and reference-mask substitutions retained the name, ID and semantic parent and preserved the unique branch and selected ID. Reference substitution produced IoU 1.0 and changed the external label from wrong-unique to correct-unique. Adding one distinct identity with the same semantic name caused ambiguity; removing the sole matching identity caused an unresolved query. This establishes a conditional causal mechanism at the tested interface: a sole matching identity is returned even when its support is wrong, because that decision has no independent spatial-quality veto. It does not establish the upstream cause of the mask error or an improvement in population performance.

The archived routing-audit decision code and frozen normalizers were executed directly. The local historical-generation helper was bound by source hash and matched all 37 archived Group routes and saved selected masks; its entire source file cannot be certified byte-identical to the original run because that helper was not included in the published frozen snapshot. Interventions used temporary or in-memory inputs and did not modify the archived packages. Reference masks were used only for diagnostic substitution and evaluation, never as an input to a learned model. These constructed resolver inputs are not new fuser outputs, and the result is specific to the evaluated benchmark interface.

S19.4 Request coverage and image error

For the imposed defect O and selected stored support M_g*, the request is U = O intersect M_g*. The evaluated image uses the backend output inside U and the corrupted input outside U. PSNR covers all of O, so omitted defect pixels in O outside U remain unfilled regardless of the generated values inside U. Every nonempty U has precision one relative to O by construction. This is not evidence of correct part identity or complete part support, and a smaller request does not itself demonstrate greater semantic precision.

The fixed LaMa checkpoint receives an image crop and request mask. The crop is determined by U's bounding box with padding equal to 22% of its longer dimension, rounded to pixels, with a minimum of 12 pixels and clipping at image boundaries. The query and Group ID are not direct network inputs. Changing U can change the mask and the image context presented to the model, even though the weights are fixed. No fixed-crop intervention separates those effects here.

TABLE S38. PSNR differences in the original 37 restoration cases stratified by paired request-recall change

HPID request recall vs raw	Cases	Mean paired PSNR change (dB)
Lower	10	-3.1423
Unchanged	22	0.0000
Higher	5	0.7860

Mean request recall was 0.4081 for raw proposals and 0.3226 for HPID. The ten cases with decreased recall averaged -3.1423 dB in HPID-minus-raw PSNR; the 22 with unchanged recall averaged 0.0000 dB; the five with increased recall averaged +0.7860 dB. These strata describe an association rather than a monotonic or causal relation. For example, laptop_computer__u01 had a recall decrease of 0.1429 and a PSNR increase of 0.4271 dB. We have not decomposed the observed PSNR difference into contributions from omitted defect pixels, mask geometry and conditional generation.

The aggregate PSNR difference is -0.7431 dB, with the original paired-bootstrap 95% interval [-1.6322, 0.0153] retained in S14/Table S27. The interval includes zero and does not establish equivalence. These observations provide no evidence of improved restoration quality and do not isolate damage to the learned generative prior. The 253-image routing and request-mask expansion does not add restoration outputs or PSNR measurements.

Table S27 retains the original 10,000-draw paired PSNR interval [-1.6322, 0.0153] (sorted case IDs; seed 20261839); the source-data recheck uses the same 37 differences in export order with seed 20261008 and yields [-1.6787, 0.0206], a finite-resampling difference rather than a new sample or changed outcome.

S19.5 Automatic roots in the original 42-object holdout

The original commands contain no reference-mask input and no explicit --asset-prompt. They operate on source crops, while the reference part masks reside separately. All 42 stored fine maps were reproduced exactly; none of the automatic root arrays equals the reference mask. The mean automatic-root IoU is 0.7287 [0.6421, 0.8076]. This corrects the manuscript wording that previously described oracle roots. It does not remove oracle crop localization.

TABLE S39. Post-review 42-case root substitution on fine Part IDs

Metric	Automatic root	Reference root	Paired change [95% CI]
Part F1@.25	0.3390	0.3713	+0.0324 [-0.0017, +0.0709]
Part F1@.50	0.1645	0.2120	+0.0474 [+0.0159, +0.0856]
Part F1@.75	0.0753	0.0870	+0.0117 [-0.0060, +0.0397]
Semantic F1@.25	0.2252	0.2437	+0.0185 [+0.0031, +0.0379]
Boundary F1@.25	0.4755	0.5249	+0.0494 [+0.0014, +0.1154]

Only the root candidate mask changes; upstream candidates are held fixed and group regrouping is not rerun. These fine-ID values therefore cannot be substituted for the original Group-ID F1 of 0.3951. The diagnostic is retrospective, with 42 paired objects as the unit. It quantifies root dependence under supplied crop localization, not full-image object discovery or a guaranteed oracle upper bound. The full-image pipeline is evaluated separately in Section S19.12.

S19.6 Controlled Group-gate removals

The named semantic, structural, and appearance predicate is invoked during Group construction, after fine-ID fusion. Each intervention starts from the same complete exported pre-Group-gate pool, including candidates rejected by the original Group predicate. We hold fine maps, Part records, source RGB, candidate masks, scores, metadata, roots, and profiles fixed, bypass only the chosen predicate decisions, and rerun actual Group construction for all eight subsets. Raw structural and appearance predicates are evaluated when earlier gates are bypassed; they retain their own semantic-source and profile inputs. Profile nomination and other assembly constraints are unchanged, so removing the semantic switch is not removal of all semantic information.

Before running the gate ablations, we verified that the archived Group implementation for each split reproduced its original outputs. The later holdout Group snapshot reproduced only 178/226 earlier test maps. The archived Group source at commit da7d236b7873200692c262d41b5f69196d9dd5fb reproduced 226/226; the holdout snapshot be543003632fa739d0b63e9acbffb8d91d99b84a reproduced 42/42. Each split uses its matching source for all eight interventions. Both pixel maps and ordered Group identifiers/semantic records match, and per-intervention input hashes are unchanged. The two splits are reported separately. Historical files and tags remain intact; the recovered source, hashes, and unsuccessful version check are retained in the evidence archive.

The 226-case gate baseline (Group F1@0.25 0.3662) replays test226_hpid, the earlier 1,951-candidate archive used in the fine-ID comparison. It is not the later v0.3.1 Group regression of 0.4043 in Table VII. That regression used 2,188 candidates from test226_identity_geometry_v12_20260824. A paired provenance audit confirms the same 226 source RGB arrays, but only 19/226 identical fine maps and 20/226 identical Group maps across these archives. The later record stating no candidate

regeneration describes regrouping relative to its own later source package, not relative to the earlier 1,951 proposals. We retain both historical results with their distinct inputs and do not pool them or imply a fixed-input comparison between them.

The complete pools contain 1,951 and 449 candidates. Profile nominations cover 1,022 candidates in 189/226 test cases and 258 in 37/42 holdout cases. All 37 and five zero-nomination cases remain in the respective case-macro denominators, with zero candidate precision/recall/F1. Candidate retention below refers to these nominations, not every exported candidate. A rejected seed can still contribute via existing fine parts or other Group assembly paths. Final Group scores use the unchanged evaluator. Pointwise intervals use 10,000 paired case-bootstrap draws, seed 20261008; seven contrasts per split are exploratory and unadjusted.

TABLE S40. Eight enabled-gate subsets, 226-case historical Group replay

Enabled	Retained seeds	Precision / recall	Group F1@.25	Paired F1 change [95% CI]
STA	546	0.3181 / 0.1977	0.3662	+0.0000 [+0.0000, +0.0000]
none	1022	0.2891 / 0.2651	0.3667	+0.0005 [-0.0124, +0.0134]
S	904	0.2892 / 0.2525	0.3665	+0.0003 [-0.0121, +0.0127]
T	615	0.3014 / 0.2047	0.3603	-0.0059 [-0.0132, +0.0003]
A	917	0.3017 / 0.2536	0.3686	+0.0024 [-0.0098, +0.0151]
ST	578	0.3044 / 0.2014	0.3635	-0.0027 [-0.0068, +0.0005]
SA	832	0.3083 / 0.2463	0.3684	+0.0022 [-0.0097, +0.0142]
TA	570	0.3159 / 0.2002	0.3640	-0.0022 [-0.0070, +0.0015]

TABLE S41. Eight enabled-gate subsets, 42-case original holdout Group replay

Enabled	Retained seeds	Precision / recall	Group F1@.25	Paired F1 change [95% CI]
STA	192	0.3313 / 0.3532	0.3951	+0.0000 [+0.0000, +0.0000]
none	258	0.2950 / 0.3898	0.3896	-0.0055 [-0.0175, +0.0070]
S	241	0.2989 / 0.3779	0.3902	-0.0049 [-0.0169, +0.0074]
T	204	0.3152 / 0.3532	0.3879	-0.0072 [-0.0157, -0.0008]
A	247	0.3045 / 0.3898	0.3939	-0.0012 [-0.0122, +0.0101]
ST	199	0.3189 / 0.3532	0.3909	-0.0042 [-0.0103, +0.0000]
SA	231	0.3127 / 0.3779	0.3944	-0.0007 [-0.0117, +0.0106]
TA	196	0.3284 / 0.3532	0.3951	+0.0000 [+0.0000, +0.0000]

S, T, and A denote semantic, structural, and appearance verification; the setting labeled none bypasses all three. With all gates enabled, Group F1@0.25 is 0.3662 on 226 cases and 0.3951 on 42. Disabling all changes it by +0.0005 [-0.0124, 0.0134] and -0.0055 [-0.0175, 0.0070]. Nomination recall falls from 0.2651 to 0.1977 and from 0.3898 to 0.3532 when the gates are enabled; precision rises from 0.2891 to 0.3181 and 0.2950 to 0.3313. At IoU 0.50, all-off minus baseline changes Group F1 by -0.0136 [-0.0286, 0.0009] and -0.0249 [-0.0549, -0.0041]. Full threshold/semantic/boundary scores and every actual predicate call are supplied in the source data. The findings show a precision-recall tradeoff, with no universal final-accuracy guarantee.

All numeric metric summaries and intervals agree across three complete runs. Two counterfactual cases, calculator__01 and scarf__03, nevertheless differ by one pixel in targeted repeated Group maps, including with a fixed Python hash seed. The cause remains unresolved. Baseline maps remain exact; counterfactual maps are not claimed to be bitwise repeatable. Any-pixel changed-map counts refer only to the saved run and are not used as stable substantive endpoints. The pixel audit and failed/verified execution logs are preserved.

S19.7 Shared-input and spatial-stage diagnosis

We used sys.settrace to read copied arrays from the unchanged frozen fuse_candidates and _hierarchical_part_ids functions. The same ordered candidate objects were then supplied to the unchanged DBSCAN implementation; input-mask hashes remained unchanged. All source/script hashes, code-line observation points, per-case statistics and 990 reference tracks are supplied. Before the first real semantic argmax, evidence arrays were copied and read with the final background convention; these four provisional argmaxes are explicitly diagnostic and are not emitted intermediate predictions. Actual semantic states and Part-ID assignment masks were recorded separately. Every stage used the same evaluator normalization and root-as-body convention. Best-reference IoU is an oracle geometry diagnostic that may select a different mask at successive stages; same-

semantic union recall divides by the same reference mask area. Reference-part counts are descriptive, not independent statistical replication. Summaries first average within each case; intervals sample the 60 categories with replacement and retain all cases within sampled categories (10,000 draws, seed 20261008). Intervals are pointwise exploratory estimates, with no multiplicity-adjusted superiority claim. For the paired endpoint diagnosis, references were joined by case identifier and reference index across the saved final HPID and DBSCAN rows. A match required the saved class-agnostic Hungarian-assigned IoU to be at least 0.25; best IoU over all predictions was used separately to test whether qualifying geometry existed. No masks were rerun, no matching was changed, and no new independent observations or confidence intervals were introduced. First and last threshold losses were determined only from the recorded actual states, excluding the four provisional diagnostic argmaxes.

The accepted-pool reference coverage was 0.634387 for any semantics and 0.251496 for matching semantics; final HPID coverage was 0.562664 and 0.147054, respectively. At the first actual semantic argmax, matching-semantic coverage was 0.130195; the subsequent direct-detail gate raised it to 0.147386 but lowered it for 63/990 reference parts. Mean best-reference IoU increased by 0.016952 (95% CI [0.010961, 0.023223]) at this gate, with 195/990 references decreasing and 389/990 increasing. Cleanup changed matching-semantic coverage by -0.000320 (95% CI [-0.000833, 0.000121]). Identity partition lowered matching-semantic coverage for 29/990 references, by -0.000052 in case-macro mean, while improving best-reference IoU on average through splitting. Two additional eligible remainder masks entered the evaluated representation and were retained separately; none was attached. Final area/sliver filtering removed 23 evaluated masks, and the full run recorded 14 visibility-sliver drops, preserved all 354 matches at IoU 0.25, and raised macro F1 from 0.310703 to 0.313311 despite small support losses. DBSCAN and HPID each emitted 1,277 evaluated masks, but DBSCAN matched 397 references versus 354 and had higher mean best-reference IoU (0.273483 versus 0.239501).

The paired geometric endpoint contains 325 references matched by both methods, 72 by DBSCAN only, 29 by HPID only, and 564 by neither. Thus the net difference of 43 matches is 72 minus 29, not a one-directional loss of 43 references. Among the 72 DBSCAN-only matches, 61 have no final HPID mask with IoU at least 0.25. The other 11 have qualifying geometry but no credited match under the saved one-to-one assignment. The assigned prediction identity was not retained in these reference tracks, so those 11 are not attributed to particular pairwise matching conflicts.

Of the 61 references lacking final HPID geometry, 53 had a qualifying accepted-pool mask. For 51, the first observed threshold loss is between the accepted pool and the first actual semantic map; for two it is at direct-detail reassignment. One of the latter subsequently recovers at cleanup and loses support again at root restoration. Eight references never have qualifying geometry in any observed actual HPID state, including the input pool. The last-loss counts are therefore 51 at the pool-to-first-map interval, one at direct-detail reassignment, and one at root restoration, with eight never supported. Of the 195 references whose best IoU decreases at direct-detail reassignment, 98 are matched by both final methods, 13 by DBSCAN only, five by HPID only, and 79 by neither. Neither 195 nor the 63 references with decreased same-semantic union coverage is a count of lost strict true positives.

TABLE S42. Spatial support through fine fusion, 226 cases and 990 reference parts

Recorded state	Best-reference IoU	Same-name union recall	Any-name union recall
input pool	0.3118	0.2515	0.6344
after same source dedup	0.3118	0.2515	0.6344
after instance cap	0.3118	0.2515	0.6344
after hierarchy duplicate filter	0.3118	0.2515	0.6344
diagnostic consensus argmax	0.1336	0.0959	0.3960
diagnostic parent support argmax	0.1287	0.0944	0.3936
diagnostic parent residual argmax	0.1512	0.0891	0.4169
diagnostic specificity argmax	0.2068	0.1238	0.5177
actual semantic argmax	0.2139	0.1302	0.5317
after direct detail gate	0.2309	0.1474	0.5641
after cleanup	0.2307	0.1471	0.5631
after root restoration	0.2308	0.1472	0.5635
identity supported before remainder	0.2396	0.1471	0.5632
identity plus eligible remainder	0.2396	0.1471	0.5632
after remainder attachment before filter	0.2396	0.1471	0.5632
final hpid	0.2395	0.1471	0.5627

Recorded state	Best-reference IoU	Same-name union recall	Any-name union recall
final dbscan	0.2735	0.1627	0.6124

Rows prefixed diagnostic are hypothetical argmax readouts of stored evidence arrays before the first actual semantic assignment, not emitted masks or independent ablated algorithms. Other rows are observed states. A reference can select a different best mask across rows. Changes are not an additive causal explanation of the DBSCAN difference. Zero replay removals applies only to the already accepted pool and says nothing about unsaved proposal-generation rejects. The candidate and per-reference tables preserve all cases, while reference counts remain descriptive rather than independent sample sizes. Strict Part F1 is a geometric one-to-one score, whereas same-semantic union coverage requires semantic agreement and permits aggregation across masks; the latter cannot substitute for the former. A pool-to-first-map transition spans several evidence and ownership operations and compares different representations. It is not an isolated ablation of any gate, and the best-overlap mask may change between states. All new subset counts are descriptive, post hoc reference-level diagnoses of this fixed benchmark.

S19.8 Executable taxonomy boundary checks

We executed 43 deterministic API fixtures against unchanged frozen code and the actual prompt-bank configuration. These synthetic code inputs test defined behaviors; they are not additional object observations or an out-of-distribution accuracy benchmark. Every assertion passed. Full inputs, returned inventories, exceptions, and source hashes are in taxonomy_behavior.json.

TABLE S43. Taxonomy and parent behavior observed in code tests

Input boundary	Observed behavior
Unmatched category string	No matching profile. Falls back to the configured default inventory or all domain parts; no new hierarchy is inferred.
Nonexistent explicit profile hint	ValueError for a profiled domain.
Completely unmatched part label	The profile label matcher returns no match.
Missing semantic/query/fallback/assembly parent	Prompt-bank construction raises ValueError. Semantic-parent cycles are also rejected.
Invalid numerical taxonomy parent	Taxonomy construction raises ValueError for an out-of-range parent index.
Unknown class with explicit taxonomy	Fusion raises ValueError; duplicate semantic names with conflicting parents also raise.
Arbitrary nonempty labels with dynamic taxonomy	Low-level fusion constructs the supplied hierarchy and can emit a labeled region. It does not check an external inventory.
Direct-source or nonempty profile provenance in Group predicate	A fixture with an unlisted name can pass. This predicate does not independently validate vocabulary membership; without semantic provenance the control fails.

Configuration validation, label matching, dynamic taxonomy construction, and Group verification are separate interfaces. Correctly formed configuration and deterministic identifiers cannot guarantee semantic correctness under a wrong profile. No unknown-category recognition, calibrated rejection, or novel-taxonomy generalization is claimed.

S19.9 Proposal-score provenance and frozen strengths

For each accepted proposal i, s_i is its stored source-dependent ranking score after any appearance, retrieval, semantic or structural refinement; it is not a common detector likelihood. We set w_i=clip[(0.45+0.55 clip(s_i,0,1)) r_i a_i,0.01,0.995], where r_i is the stored source/profile reliability multiplier and a_i is the optional cross-family agreement factor. The released agreement setting gives a_i=1. Profile priorities can make r_i exceed 1; the bound applies to w_i. A separate broad-scene fallback multiplies the already clipped weight by 0.72 and clips again, although none of the 1,951 benchmark proposals invoked this branch. These fixed transformations shape competing evidence rather than calibrate correctness probabilities. The implementation takes a maximum within each semantic/family cell and a noisy-OR across recorded family cells; dependence between shared model stages remains a limitation.

We reconstructed the full released Eq. (2) arrays before parent support for all 226 archived cases, requiring bit equality. For a controlled stress test, the first eligible stored reference per case supplied at least 20 pixels outside differently named non-body annotations. Both the reference class and a different nonroot class had to exist in the archived candidate-derived taxonomy; the lexicographically first eligible incorrect class was selected. No candidate score, predicted-mask overlap or stress outcome selected the target. This admitted 150 cases in 51 categories and excluded 76 with recorded reasons. The incorrect mask used this annotated region, score 1 and r = 0.5, 0.75 or 1, with the original soft membership and clipping. At the consensus boundary, we injected one copy, five identical copies under one recorded family key, or five copies under five recorded keys.

The base key was the first recorded family producing the incorrect class; the other four were the first distinct keys in the global archived list. Family-max/noisy-OR, ungrouped candidate noisy-OR, all-proposal maximum and family-max/noisy-OR with uniform r each used its own uninjected baseline. Uniform r = 1 applied to original and injected proposals, making injected w = 0.995 at every nominal r level in that variant. Readouts were wrong-versus-correct evidence dominance, nonroot top-class fractions and mean evidence; normalized aliases were pooled by maximum and argmax ties used the smallest taxonomy index. These are annotation-defined conflicts and Eq. (2) readouts, not final predictions or estimated error prevalence.

The nine recorded families comprise 226 detector-derived root candidates and 32 retrieved detector candidates; 474 detector-derived profile refinements; 39 CLIPSeg-SAM2 dense supplements; 1067 SAM2 visual proposals; 42 appearance contours; 53 repetitive details; 2 axial residuals; 3 axial partitions; and 13 topology proposals. A family label is the explicit metadata grouping key, not necessarily the producing network: profile-dense retains a grounded-family label but its source is CLIPSeg-SAM2, and topology source v2 intentionally retains family v1. Grounding DINO supplies a maximum sigmoid token score; CLIPSeg supplements use the regional maximum of a smoothed sigmoid heatmap; SAM2 visual proposals begin with predicted mask IoU, then may undergo view/crop score scaling. Contour, repetitive, structural and topology scores use the documented deterministic formulas. Every accepted visual/contour proposal is subsequently scored as clip(0.62 s0+0.38 u,0,1); semantic reranking can blend this with an inventory-relative softmax weight. The metadata field sam_quality in visual proposals holds the pre-appearance proposal score, including contour heuristics, so it is not universally a raw SAM prediction. Full producer/transform formulas and source excerpts are in score_source_rules.json.

TABLE S44. Observed stored score, reliability and effective-weight ranges

Recorded family / producer	n	Stored score s	Reliability r	Effective w
Grounded roots/retrieved	258	0.2163 to 0.9526	0.6564 to 0.9979	0.3779 to 0.9606
Profile-dense (CLIPSeg)	39	0.0434 to 0.9257	0.2779 to 0.7356	0.1317 to 0.6650
Profile-refine (detector)	474	0.2402 to 0.6911	0.6610 to 1.1366	0.4118 to 0.7790
Appearance contour	42	0.4038 to 0.8067	0.6200 to 0.6800	0.4436 to 0.6077
Repetitive detail	53	0.5201 to 0.9400	0.7600 to 0.7600	0.5594 to 0.7349
Structural v1	2	0.8548 to 0.8959	0.7400 to 0.7400	0.6809 to 0.6976
Structural v2	3	0.8659 to 0.8933	0.7400 to 0.7400	0.6854 to 0.6966
Topology family v1	13	0.3120 to 0.6188	0.8200 to 0.8200	0.5097 to 0.6481
SAM2 visual family	1067	0.4566 to 0.9528	0.5687 to 0.8200	0.4517 to 0.6873

Across the frozen pool, s ranged from 0.0433852337 to 0.9528129163, r from 0.2778563447 to 1.1365760742, a was 1 for all 1951 proposals, and the final effective w ranged from 0.1316655295 to 0.9606005189. Ten reliability multipliers exceeded 1, all in profile refinement. All 1951 score/reliability/agreement/weight records exactly matched the earlier fixed-candidate sensitivity audit. Per-family ranges and exact counts are supplied in score_family_ranges.csv; these are descriptive properties of the stored pool.

Retained metadata reproduced 1213 stored score values with no discrepancy above 1e-10. The remaining 738 comprise 732 detector-origin chains whose original logits/pre-retrieval scores were not re-inferred, four cross-source maximum operations whose other precursor score is not retained in the incumbent metadata, and two topology candidates whose anchors are absent from the exported pool. Reliability formulas reproduce 1944/1951 final values. For seven profile-resolved roots, profile_resolution.py retains the geometry root sam_quality but takes the maximum reliability over broad/evidence/geometry roots; the final stored value therefore cannot be recovered from that single q field. These seven simple-formula exceptions are preserved in score_reconstruction_exceptions.csv, and archived final r values are used unchanged. The missing intermediate records limit reconstruction of these values; we therefore retained the archived reliability values unchanged. The additional empirical diagnostics below do not rerun upstream models or fit a probability-calibration function.

For the annotation-conditional reliability diagnostic, the original root/body rule retained 1,830 of 1,951 candidates. Of these, 892 candidates from 172 cases and 56 categories had same-semantic annotated references; the remaining 938 retained missing semantic outcomes. Candidate success meant maximum same-semantic IoU at least 0.25, without one-to-one assignment. Mean w was 0.5564 and success was 318/892 (0.3565). Under the hypothetical probability interpretation, Brier = mean((w-y)^2) was 0.2527 and ECE, weighted across ten fixed equal-width bins on [0,1], was 0.1999. At IoU 0.50 the success count was 202/892, Brier 0.2731 and ECE 0.3300. Estimates are candidate-weighted; repeated or overlapping proposals are not independent replicates. Missing same-class annotation does not identify a false positive. Family counts, empty bins, sparse-support flags and pointwise intervals are retained in family_empirical_reliability.csv and fixed10bin_empirical_reliability.csv. No weight or probability-calibration function was fitted.

In all 150 eligible stress cases, same-family repetition left the released evidence unchanged at every tested strength. In ungrouped noisy-OR, five copies minus one increased the case-mean wrong-top-nonroot fraction by 0.4098 [0.3437, 0.4761], 0.1229 [0.0768, 0.1735] and 0.0241 [0.0098, 0.0421] for r = 0.5, 0.75 and 1, respectively. These are separate within-method

repetition increments. For the release at r = 0.75, that fraction was 0.2714 without injection, 0.8614 with one mask, 0.8614 with five same-family copies and 1.0000 with five cross-family copies. Thus within-family maximum prevents additional duplicate accumulation, while neither the weight nor the grouping blocks a strong single error or errors repeated across grouping keys. The keys do not imply independent networks. All modes, strengths and endpoints, including each method's clean baseline, are retained in consensus_stress/summary.csv and paired_comparisons.csv.

The separate clean full-fusion replay used every archived case and the original evaluator. Released maps were bit-identical in all 226 cases. Released, no-consensus (cross-family maximum) and uniform-r Part F1@0.25 values were 0.3133, 0.3019 and 0.3321; semantic F1@0.25 values were 0.1396, 0.1332 and 0.1424. Release minus no-consensus was +0.0114 [0.0059, 0.0174] for Part F1 and +0.0064 [0.0006, 0.0127] for semantic F1. Uniform r minus release was +0.0187 [0.0040, 0.0346] and +0.0028 [-0.0077, 0.0127], respectively. Uniform r substitutes reliability throughout fusion, including filtering and ownership, rather than isolating Eq. (2). All five metrics and both contrasts are retained in clean_ablation/summary.csv and paired_deltas.csv. These results do not establish optimality of the frozen reliability values.

The stress and clean-fusion analyses use case-macro means and paired differences; the reliability diagnostic uses candidate-weighted ratios. All use 10,000 category-cluster bootstrap draws (seed 20261010), sampling the original 60 categories with replacement and retaining their eligible observations and pairing. Stress eligibility is conditional on the archived taxonomy. Empty denominators are omitted only where estimates are undefined and are counted in the reliability tables. Intervals are pointwise exploratory 95% estimates without multiplicity-adjusted superiority claims. The studies reuse an evaluated accepted-candidate benchmark, not untouched validation data; upstream models were not rerun and no threshold or reliability value was selected from these results.

The frozen fusion SHA256 matches the recorded frozen-source checksum and the used prompt-bank SHA256 matches the benchmark manifest. All three used prompt configuration JSON files are semantically and newline-normalized-text identical to the snapshot versions; their raw byte hashes differ because line endings differ, and both are recorded. Source, dependency and input JSON hashes are listed. This verifies the current frozen inputs/rules and preserves them; it does not independently date the original freezing of every model checkpoint.

S19.10 Original 37-case review-option audit

A review option is one exported part instance returned for a semantic request. Mean option count averages these returned choices over all 37 requests. We replayed five frozen methods and reproduced all 185 original method–request endpoints to tolerance 1e-12. The requests correspond to 37 distinct source images. All methods use the same exposure rule, returning normalized-semantic matches when present and otherwise all exported instances. No output or review set was empty. This post-hoc audit supplements the previously reported means; it changes no method, request, mask, or parameter. This section retains the earlier 37-case audit. The expanded protocol and all-cohort comparisons are in S19.13.

For option counts, 10,000 paired case-bootstrap draws (seed 20261009) retain all methods within each sampled request. Table S45 reports 98.75% individual percentile intervals, giving nominal Bonferroni familywise 95% coverage over the four HPID-minus-baseline comparisons. Relative reductions are ratios of cohort means. Routing and availability are distinct: a semantic success requires a matching name and target IoU ≥0.25; exposed-set geometric availability also counts correctly overlapping fallback instances whose names may be wrong. All count denominators are 37.

TABLE S45. Historical mean review options and correctness (original 37 requests)

Method	Mean options	Unique correct	Semantic correct	Exposed geometry	Wrong unique	HPID minus method Difference [98.75% CI]
Raw proposals	3.6757	9	18	22	5	−1.6216 [−2.8378, −0.5405]
Greedy NMS	3.7838	9	11	21	4	−1.7297 [−3.3243, −0.5269]
DBSCAN	2.8108	14	15	21	5	−0.7568 [−1.7704, 0.0677]
Local pairwise CRF	4.3243	6	13	20	3	−2.2703 [−4.1081, −0.7568]
HPID	2.0541	14	15	19	6	—

HPID returned 2.0541 review options per request, compared with 3.7838 for NMS, 3.6757 for raw proposals, and 4.3243 for local pairwise CRF. The corresponding reductions were 45.7%, 44.1%, and 52.5%; all three adjusted intervals in Table S45 excluded zero. The following correctness comparison uses NMS as the baseline. HPID's observed uniquely correct and semantically available counts were 14 versus 9 and 15 versus 11, respectively; all NMS successes in these two endpoints were retained. These remain descriptive binary comparisons: gains/losses were 5/0 and 4/0, with exact two-sided McNemar p=0.0625 and p=0.125. Actual exposed-set geometric availability was 19 versus 21 (2 gains, 4 losses), so fewer options cannot be interpreted as lossless pruning or demonstrated human-time savings.

The DBSCAN mean was 2.8108 versus 2.0541, a descriptive 26.9% reduction; the four-comparison-adjusted interval includes zero. Both methods had 14 uniquely correct and 15 semantically available targets, with one success exchanged in each endpoint. On the 14 requests where both methods exposed a semantically correct target, their option counts were identical. The full-cohort reductions in returned options relative to raw proposals and local pairwise CRF are supported by the adjusted intervals in Table S45. Complete IoU 0.50 results, paired changes, exact binary diagnostics, and all prespecified auxiliary subsets are retained in the accompanying evidence records. These data do not establish statistical equivalence of accuracy or a measured reduction in operator effort.

S19.11 Correct identity retention under controlled input changes

A separate locally fixed, post-hoc protocol reran HPID, DBSCAN, and greedy NMS on the existing 226-image, 1,951-candidate cohort. Nine conditions comprise root erosion, dilation and translation as in S15; nonroot scores multiplied independently by U(0.95,1.05) and then clipped to [0,1]; and removal of round(0.10N) nonroot candidates. Score and removal conditions each use seeds 20261009–20261011. The removal rule leaves 94/226 cases unchanged because of integer rounding. All methods receive identical perturbed inputs. Proposal generation is not rerun.

A shared adapter assigns each nonempty output a semantic/rank key by sorting within its raw semantic name on centroid y, centroid x, negative area, and mask digest. Ground truth is used only for semantic one-to-one IoU scoring, never to assign IDs. The primary cohort contains 126 targets initially correct for all three methods at IoU 0.25, in 90/226 cases. A target is retained only when the same key still matches the same reference correctly. We average the three conditions per family, the three families, and then eligible cases equally. The 136 cases without a common initially correct target are excluded only from this conditional endpoint. The all-reference column uses all 990 reference targets: it requires initial and perturbed correctness with the same key, divides by the full reference count in each case, then averages the 226 case rates equally. All rates are case-macro averages.

TABLE S46. Correct identity retention and full-cohort quality

Method	Correct same-key retention Common initial targets	Correct same-key retention All reference targets	Perturbed Part F1@0.25
HPID	93.99%	14.91%	0.3049
DBSCAN	96.01%	16.71%	0.3492
NMS	93.98%	15.53%	0.3364

The primary HPID-minus-DBSCAN difference was −2.02 percentage points (97.5% interval [−4.01, −0.08]); against NMS it was +0.01 [−2.63, 2.65]. These paired 10,000-draw intervals (seed 20261009) adjust for two primary comparisons. Category-cluster sensitivity for DBSCAN crossed zero. Native-ID retention was 91.30%, 92.98%, and 91.44%, respectively, and did not establish an advantage. The 6,780 predictions had no execution failures; all 226 original HPID maps and published baseline metrics were reproduced. Full conditions, IoU 0.50, family-adjusted comparisons, and alternative weighting are retained in the evidence archive. This audit concerns controlled changes to existing inputs, not a fresh holdout, image-level occlusion recovery, or video tracking.

S19.12 Automatic primary-object and Group evaluation on full images

This retrospective paired extension includes all 42 actually materialized holdout objects from 42 distinct source images. The protocol was fixed locally before the new runs, without public preregistration. Complete source images retain their archived dimension normalization and are copied to anonymous input paths. Inference consumes the image, frozen models and general configuration, without an annotated box, root mask, category, object identifier or target point. The v0.3.1 snapshot be54300 and all seven historically frozen assets remain unchanged. The 42 commands change only the image and output paths. The original primary-object policy selects the output autonomously. Wrong-object selections, empty outputs and failures remain in the 42-case denominator; no root is reselected using ground truth.

An environment control reruns the first archived crop and exactly reproduces its Part and Group pixel maps and ordered identity/semantic records. Historical Group F1@0.25 is reproduced at 0.3951099151 across all 42 cases. Translating cropped predictions into full-image coordinates preserves every object and part reference pixel, without truncation. Scoring retains the released maximum-total-IoU Hungarian assignment followed by thresholding and the original root/body inclusion rule. All applicable exported Group and Part records enter scoring, without additional reference-based category or spatial filtering. Root metrics use the union of exported root candidates; the root box is its tight bounding box. The final exported foreground union is evaluated separately.

The primary endpoint is all-case macro Group F1@0.25. The full-image estimate is 0.2532, versus cropped 0.3951, with paired difference -0.1419 [-0.2378, -0.0552]. Cases are sorted by case_id and resampled as paired source images in 10,000 bootstrap draws, seed 20261009. Intervals in Tables S47-S48 are percentile 95% intervals. Secondary intervals are descriptive and unadjusted for multiplicity. The run produces 42/42 valid packages. Mean full-image time is 33.36 seconds, including process and model startup.

TABLE S47 Paired segmentation and root results for full-image and cropped inputs

Metric	Crop	Full image	Full minus crop [95% interval]
Group F1@0.25	0.3951	0.2532	-0.1419 [-0.2378, -0.0552]
Group F1@0.50	0.2417	0.1433	-0.0984 [-0.1904, -0.0149]
Group F1@0.75	0.1035	0.0792	-0.0243 [-0.0925, +0.0381]
Part F1@0.25	0.3390	0.2323	-0.1066 [-0.1839, -0.0337]
Part F1@0.50	0.1645	0.1411	-0.0234 [-0.0901, +0.0447]
Part F1@0.75	0.0753	0.0775	+0.0021 [-0.0484, +0.0527]
Root mask IoU	0.7287	0.4600	-0.2686 [-0.3855, -0.1606]
Root bounding-box IoU	0.8361	0.5393	-0.2969 [-0.4210, -0.1810]
Exported foreground IoU	0.7283	0.4599	-0.2684 [-0.3853, -0.1604]

TABLE S48 Automatic selection and output validity

Endpoint	Crop	Full image	Proportion difference [95% interval]
Root IoU >= 0.25	38/42	22/42	-0.3810 [-0.5238, -0.2381]
Root IoU >= 0.50	34/42	22/42	-0.2857 [-0.4286, -0.1429]
Correct domain	40/42	31/42	-0.2143 [-0.3333, -0.0952]
Correct profile	39/42	26/42	-0.3095 [-0.4524, -0.1667]
Valid package	42/42	42/42	+0.0000 [+0.0000, +0.0000]

Full-image processing exports a mean of 4.4048 Groups and 6.1905 Parts per case. After the released body-inclusion rule, the mean scored counts are 4.4048 and 5.7143, respectively. Each mean includes all 42 cases.

Root-overlap success in Table S48 is measured against the originally fixed target, not all-object scene detection AP. Domain and profile scores use the frozen category configuration of the actually materialized object; four historical cross-category replacement slots use their replacement profiles, while stale slot labels never enter inference. Removing the annotated crop jointly changes target selection, image context/scale and subsequent root/part generation, so the contrast cannot isolate detector error. These full images correspond to the original holdout objects; the extension does not constitute a new independent untouched test. Per-case outcomes, complete commands, input/output hashes and scoring scripts are retained in the full_image_review4 evidence directory.

S19.13 Expanded routing and controlled request-mask evaluation on 253 image-level requests

The expanded cohort combines the original 37 requests with 216 eligible images from the existing 226-image test archive. Source image identifiers and image hashes confirm 253 distinct images and no overlap between cohorts. One target is fixed per image. The unchanged rule requires at least 24 target pixels and 4%–75% of object area, then ranks targets by distance from 25% of object area, larger area and name. Ten of the 226 images have no eligible target. The original requests, defect seeds and masks are retained. Reference annotations fix eligibility and the target before routing; target geometry defines the controlled defect and scoring but is withheld from semantic resolution. This is retrospective reuse of existing benchmarks, not a new untouched test.

The two cohorts retain their different archived upstream proposal pools and prompt-bank histories. Each case supplies identical candidates to raw proposals, greedy NMS, DBSCAN, local pairwise CRF and HPID. HPID Part maps reproduce the archive; Group maps are reconstructed for all 253 requests with the common frozen be54300 implementation, and all 37 original Group outputs are reproduced exactly. No foundation or inpainting model is rerun and no parameter is retuned. All 1,265 method/request runs complete, with no empty exposed sets. Thus a failure or empty output cannot explain the observed option reduction. The original 37-case LaMa quality and repeatability results in S14/S19.4 remain separate.

A review option is an exported instance exposed under the same rule for every method: use normalized semantic matches if any exist, otherwise expose all exported instances. The mean includes all requests, not only ambiguous requests. At each target-IoU threshold, exactly one match is either correct-unique or wrong-unique; multiple matches are ambiguous and zero matches unresolved. These four states partition every cohort. Semantic availability requires any matching option to reach the threshold. Exposed-set geometric availability also counts the fallback set; availability anywhere in the export is recorded separately. These measures are distinct from request recall, which is the fraction of controlled defect covered by the selected request mask, and from semantic precision. Counts are not human time.

The protocol, request manifest and analysis implementation were fixed before expanded routing execution. The source image is the paired unit. For the combined analysis, each of 10,000 bootstrap draws (NumPy default_rng, seed 20261009) samples 37 original and 216 additional requests with replacement within their cohorts, preserving all methods for each request. Four primary HPID-minus-baseline option contrasts use 98.75% percentile intervals, a Bonferroni adjustment for nominal 95% family coverage. Relative reductions use cohort means. Separate-cohort contrasts are descriptive four-comparison families. Table S45 retains its earlier bootstrap realization; Table S49 uses this expanded protocol's fixed request order and random-number sequence, with identical original-37 endpoints. Binary proportions have pointwise Wilson 95% intervals; paired binary differences and exact McNemar diagnostics are descriptive, not evidence of equivalence.

Mean options for HPID/raw/NMS/DBSCAN/CRF are 2.05/3.68/3.78/2.81/4.32 in the original 37, 3.09/4.70/3.98/3.89/5.64 in the additional 216, and 2.94/4.55/3.95/3.74/5.45 overall. All four adjusted contrasts exclude zero in the combined and additional cohorts. The original-37 DBSCAN interval crosses zero. The expanded evidence supports fewer exposed choices under this rule; it does not retrospectively change the uncertainty of the earlier 37-case comparison.

TABLE S49. Paired review-option comparisons by cohort

Cohort	Comparator	HPID mean	Baseline mean	Difference [98.75% CI]	Fewer (%)
Original 37	Raw proposals	2.0541	3.6757	-1.6216 [-2.8649, -0.5405]	44.12%
Original 37	Greedy NMS	2.0541	3.7838	-1.7297 [-3.2973, -0.5135]	45.71%
Original 37	DBSCAN	2.0541	2.8108	-0.7568 [-1.8378, +0.0677]	26.92%
Original 37	Local pairwise CRF	2.0541	4.3243	-2.2703 [-4.1218, -0.7297]	52.50%
Additional 216	Raw proposals	3.0880	4.7037	-1.6157 [-2.2824, -0.9815]	34.35%
Additional 216	Greedy NMS	3.0880	3.9769	-0.8889 [-1.3264, -0.4954]	22.35%
Additional 216	DBSCAN	3.0880	3.8935	-0.8056 [-1.2639, -0.3657]	20.69%
Additional 216	Local pairwise CRF	3.0880	5.6435	-2.5556 [-3.4838, -1.7315]	45.28%
Combined 253	Raw proposals	2.9368	4.5534	-1.6166 [-2.1917, -1.0415]	35.50%

Cohort	Comparator	HPID mean	Baseline mean	Difference [98.75% CI]	Fewer (%)
Combined 253	Greedy NMS	2.9368	3.9486	-1.0119 [-1.4348, -0.6245]	25.63%
Combined 253	DBSCAN	2.9368	3.7352	-0.7984 [-1.2174, -0.3992]	21.38%
Combined 253	Local pairwise CRF	2.9368	5.4506	-2.5138 [-3.3281, -1.7727]	46.12%

Options per request. Difference = HPID minus comparator. Intervals are 98.75% percentile intervals, adjusted within four contrasts; negative values favor fewer HPID options. The original 37 requests and reused 216 images form separate archival cohorts.

At IoU 0.25, HPID's original-37 state counts are 14 correct-unique, 6 wrong-unique, 6 ambiguous and 11 unresolved; additional-216 counts are 19, 30, 24 and 143; combined counts are 33, 36, 30 and 154. The original wrong-unique incidence is 6/37 (16.22%; 95% CI [7.65%, 31.14%]); the combined incidence is 36/253 (14.23%; [10.46%, 19.07%]). Wrong among unique is a different conditional risk: 6/20 (30.00%; [14.55%, 51.90%]) originally, 30/49 (61.22%; [47.25%, 73.57%]) additionally, and 36/69 (52.17%; [40.59%, 63.53%]) overall. Raw has 39/77 (50.65%; [39.72%, 61.52%]) overall. Fewer ambiguous outputs do not establish safer unique returns. At IoU 0.50 the combined HPID wrong-among-unique rate is 52/69 (75.36%; [64.04%, 84.01%]).

TABLE S50. All routing states by cohort, method and threshold

Cohort	Method	IoU	Correct unique	Wrong unique	Ambiguous	Unresolved
Original 37	Raw proposals	.25	9/37 0.243 [0.134, 0.401]	5/37 0.135 [0.059, 0.280]	13/37 0.351 [0.218, 0.512]	10/37 0.270 [0.154, 0.430]
Original 37	Greedy NMS	.25	9/37 0.243 [0.134, 0.401]	4/37 0.108 [0.043, 0.247]	8/37 0.216 [0.114, 0.372]	16/37 0.432 [0.287, 0.591]
Original 37	DBSCAN	.25	14/37 0.378 [0.241, 0.539]	5/37 0.135 [0.059, 0.280]	7/37 0.189 [0.095, 0.342]	11/37 0.297 [0.175, 0.458]
Original 37	Local pairwise CRF	.25	6/37 0.162 [0.077, 0.311]	3/37 0.081 [0.028, 0.213]	14/37 0.378 [0.241, 0.539]	14/37 0.378 [0.241, 0.539]
Original 37	HPID Group IDs	.25	14/37 0.378 [0.241, 0.539]	6/37 0.162 [0.077, 0.311]	6/37 0.162 [0.077, 0.311]	11/37 0.297 [0.175, 0.458]
Original 37	Raw proposals	.50	2/37 0.054 [0.015, 0.177]	12/37 0.324 [0.196, 0.485]	13/37 0.351 [0.218, 0.512]	10/37 0.270 [0.154, 0.430]
Original 37	Greedy NMS	.50	5/37 0.135 [0.059, 0.280]	8/37 0.216 [0.114, 0.372]	8/37 0.216 [0.114, 0.372]	16/37 0.432 [0.287, 0.591]
Original 37	DBSCAN	.50	6/37 0.162 [0.077, 0.311]	13/37 0.351 [0.218, 0.512]	7/37 0.189 [0.095, 0.342]	11/37 0.297 [0.175, 0.458]
Original 37	Local pairwise CRF	.50	3/37 0.081 [0.028, 0.213]	6/37 0.162 [0.077, 0.311]	14/37 0.378 [0.241, 0.539]	14/37 0.378 [0.241, 0.539]

Cohort	Method	IoU	Correct unique	Wrong unique	Ambiguous	Unresolved
Original 37	HPID Group IDs	.50	7/37 0.189 [0.095, 0.342]	13/37 0.351 [0.218, 0.512]	6/37 0.162 [0.077, 0.311]	11/37 0.297 [0.175, 0.458]
Additional 216	Raw proposals	.25	29/216 0.134 [0.095, 0.186]	34/216 0.157 [0.115, 0.212]	44/216 0.204 [0.155, 0.262]	109/216 0.505 [0.438, 0.571]
Additional 216	Greedy NMS	.25	25/216 0.116 [0.080, 0.165]	34/216 0.157 [0.115, 0.212]	26/216 0.120 [0.083, 0.171]	131/216 0.606 [0.540, 0.669]
Additional 216	DBSCAN	.25	25/216 0.116 [0.080, 0.165]	40/216 0.185 [0.139, 0.242]	28/216 0.130 [0.091, 0.181]	123/216 0.569 [0.503, 0.634]
Additional 216	Local pairwise CRF	.25	21/216 0.097 [0.064, 0.144]	33/216 0.153 [0.111, 0.207]	31/216 0.144 [0.103, 0.197]	131/216 0.606 [0.540, 0.669]
Additional 216	HPID Group IDs	.25	19/216 0.088 [0.057, 0.133]	30/216 0.139 [0.099, 0.191]	24/216 0.111 [0.076, 0.160]	143/216 0.662 [0.597, 0.722]
Additional 216	Raw proposals	.50	17/216 0.079 [0.050, 0.122]	46/216 0.213 [0.164, 0.272]	44/216 0.204 [0.155, 0.262]	109/216 0.505 [0.438, 0.571]
Additional 216	Greedy NMS	.50	16/216 0.074 [0.046, 0.117]	43/216 0.199 [0.151, 0.257]	26/216 0.120 [0.083, 0.171]	131/216 0.606 [0.540, 0.669]
Additional 216	DBSCAN	.50	17/216 0.079 [0.050, 0.122]	48/216 0.222 [0.172, 0.282]	28/216 0.130 [0.091, 0.181]	123/216 0.569 [0.503, 0.634]
Additional 216	Local pairwise CRF	.50	13/216 0.060 [0.036, 0.100]	41/216 0.190 [0.143, 0.247]	31/216 0.144 [0.103, 0.197]	131/216 0.606 [0.540, 0.669]
Additional 216	HPID Group IDs	.50	10/216 0.046 [0.025, 0.083]	39/216 0.181 [0.135, 0.237]	24/216 0.111 [0.076, 0.160]	143/216 0.662 [0.597, 0.722]
Combined 253	Raw proposals	.25	38/253 0.150 [0.111, 0.199]	39/253 0.154 [0.115, 0.204]	57/253 0.225 [0.178, 0.281]	119/253 0.470 [0.410, 0.532]
Combined 253	Greedy NMS	.25	34/253 0.134 [0.098, 0.182]	38/253 0.150 [0.111, 0.199]	34/253 0.134 [0.098, 0.182]	147/253 0.581 [0.519, 0.640]
Combined 253	DBSCAN	.25	39/253 0.154 [0.115, 0.204]	45/253 0.178 [0.136, 0.230]	35/253 0.138 [0.101, 0.186]	134/253 0.530 [0.468, 0.590]
Combined 253	Local pairwise CRF	.25	27/253 0.107 [0.074, 0.151]	36/253 0.142 [0.105, 0.191]	45/253 0.178 [0.136, 0.230]	145/253 0.573 [0.512, 0.633]

Cohort	Method	IoU	Correct unique	Wrong unique	Ambiguous	Unresolved
Combined 253	HPID Group IDs	.25	33/253 0.130 [0.094, 0.178]	36/253 0.142 [0.105, 0.191]	30/253 0.119 [0.084, 0.164]	154/253 0.609 [0.547, 0.667]
Combined 253	Raw proposals	.50	19/253 0.075 [0.049, 0.114]	58/253 0.229 [0.182, 0.285]	57/253 0.225 [0.178, 0.281]	119/253 0.470 [0.410, 0.532]
Combined 253	Greedy NMS	.50	21/253 0.083 [0.055, 0.124]	51/253 0.202 [0.157, 0.255]	34/253 0.134 [0.098, 0.182]	147/253 0.581 [0.519, 0.640]
Combined 253	DBSCAN	.50	23/253 0.091 [0.061, 0.133]	61/253 0.241 [0.193, 0.297]	35/253 0.138 [0.101, 0.186]	134/253 0.530 [0.468, 0.590]
Combined 253	Local pairwise CRF	.50	16/253 0.063 [0.039, 0.100]	47/253 0.186 [0.143, 0.238]	45/253 0.178 [0.136, 0.230]	145/253 0.573 [0.512, 0.633]
Combined 253	HPID Group IDs	.50	17/253 0.067 [0.042, 0.105]	52/253 0.206 [0.160, 0.260]	30/253 0.119 [0.084, 0.164]	154/253 0.609 [0.547, 0.667]

Each entry gives k/n and proportion [pointwise 95% Wilson CI]. The four state counts sum to n; ambiguity and unresolved states depend on match count and therefore do not vary with the scoring threshold.

The original correct-in-review proportions 9/13 for raw and 1/6 for HPID have different ambiguous subsets. Tracking the fixed nine raw-ambiguous requests containing a correct option gives five correct-unique, one still ambiguous with a correct option, two wrong-unique and one ambiguous without a correct option. Six retain a correct semantic option and three lose it; this is not loss of eight requests. In the additional cohort, raw 24/44 and HPID 14/24 arise from six correct-unique, fourteen correct-ambiguous, one incorrect-ambiguous and three unresolved outcomes among the fixed 24 originally correct ambiguous requests. One of the three unresolved requests still has correct geometry in the fallback set. The combined conditional counts are raw 33/57 and HPID 15/30. Complete transition matrices retain both IoU thresholds and all methods.

TABLE S51. Conditional review correctness and error among unique outputs

Cohort	IoU	Raw correct / ambiguous	HPID correct / ambiguous	Raw wrong / unique	HPID wrong / unique
Original 37	.25	9/13 0.692 [0.424, 0.873]	1/6 0.167 [0.030, 0.564]	5/14 0.357 [0.163, 0.612]	6/20 0.300 [0.145, 0.519]
Original 37	.50	7/13 0.538 [0.291, 0.768]	0/6 0.000 [0.000, 0.390]	12/14 0.857 [0.601, 0.960]	13/20 0.650 [0.433, 0.819]
Additional 216	.25	24/44 0.545 [0.401, 0.683]	14/24 0.583 [0.388, 0.755]	34/63 0.540 [0.418, 0.657]	30/49 0.612 [0.472, 0.736]
Additional 216	.50	15/44 0.341 [0.219, 0.489]	6/24 0.250 [0.120, 0.449]	46/63 0.730 [0.610, 0.824]	39/49 0.796 [0.664, 0.885]
Combined 253	.25	33/57 0.579 [0.450, 0.698]	15/30 0.500 [0.332, 0.668]	39/77 0.506 [0.397, 0.615]	36/69 0.522 [0.406, 0.635]
Combined 253	.50	22/57 0.386 [0.271, 0.516]	6/30 0.200 [0.095, 0.373]	58/77 0.753 [0.646, 0.836]	52/69 0.754 [0.640, 0.840]

Entries give k/n and proportion [pointwise 95% Wilson CI]. Ambiguous and unique denominators differ by method and must not be interpreted as a fixed paired subset; fixed-subset transitions are described in the text.

At IoU 0.25, correct semantic availability changes from raw 18 to HPID 15 in the original cohort (14 retained, 4 lost, 1 gained), and from 53 to 33 in the additional cohort (32 retained, 21 lost, 1 gained). Overall, 71 to 48 decomposes into 46 retained, 25 lost, 2 gained and 180 absent under both methods. The paired availability difference is −9.09 percentage points

(descriptive 95% interval [−13.04, −5.53]). These 25 lost-availability requests are not the 36 wrong-unique results; the latter are defined by the final route state. Exposed-set geometric availability is raw 124 versus HPID 111, with 98 retained, 26 lost and 13 gained. At IoU 0.50, semantic availability is 41 versus 23, with 21 retained, 20 lost and 2 gained. The complete paired guards also report availability anywhere in the export and both-correct subsets.

All 25 primary-threshold losses retain a geometrically correct, semantically matching raw candidate in the fuser's accepted list. The first observed loss is between accepted candidates and native Part export in 24 cases, and between Part and Group in one. Of the 24 earlier losses, 17 retain correct individual-Part geometry under another name and seven have no individual Part reaching IoU 0.25. At final Group export, 15/25 retain correct geometry under another name. The single Part-to-Group loss is laptop_computer__01: base-panel IoU falls from raw 0.7184 to Part 0.2998, then no matching Group; all 6,799 pixels of that Part enter a bezel Group. These observations distinguish spatial support, semantic assignment and Group merging. They do not isolate direct evidence gates, ownership competition, instance extraction and cleanup within the accepted-to-Part stage. A rejection count alone is not a case-specific cause. All four original-cohort losses and all 20 secondary-threshold losses remain in the case records, including the box, car, scissors and shoe traces in S19.3.

The combined 36 wrong-unique outcomes occur across all six domains, including 12/80 container requests. Daily objects have 28/37 unresolved requests and devices 37/66. The 61 actual object categories contain only 1–7 images each. For example, bucket has 3/4 wrong-unique requests and knife 2/7. These sparse counts and their wide intervals are descriptive, not a category ranking or evidence that zero-observed-error categories are reliable. Table S52 shows domain denominators; domain/category counts, confidence intervals and all wrong-unique records for every cohort, method and threshold are retained in the source data.

TABLE S52. Combined-cohort HPID failures by domain

Domain	Images	Wrong unique at IoU .25	Unresolved
container	80	12/80 0.150 [0.088, 0.244]	48/80 0.600 [0.490, 0.700]
daily_object	37	7/37 0.189 [0.095, 0.342]	28/37 0.757 [0.599, 0.866]
device	66	6/66 0.091 [0.042, 0.184]	37/66 0.561 [0.441, 0.674]
furniture	20	3/20 0.150 [0.052, 0.360]	13/20 0.650 [0.433, 0.819]
tool_prop	38	6/38 0.158 [0.074, 0.304]	23/38 0.605 [0.447, 0.744]
vehicle	12	2/12 0.167 [0.047, 0.448]	5/12 0.417 [0.193, 0.680]

Entries give k/n and proportion [pointwise 95% Wilson CI]. The six domains sum to 253 images. All 61 category strata and all methods/thresholds are retained in domain_category_failure_summary.csv; small strata do not support category-specific superiority claims.

The handoff_review6 archive contains the frozen protocol, request manifest, implementation hashes, routing_cases.csv, all 1,265 request-mask hashes and case traces. Statistics retain the four option contrasts, all route states, conditional rates, paired availability transitions, exact binary diagnostics, domain/category strata and failures. The mechanism audit retains all 253 stage-availability records and all 30 requests with a loss at either threshold. Independent audits reproduce all 185 original method/request endpoints and verify 253 unique source images, unchanged inputs and the complete mechanism counts. The expansion supplies no new PSNR, LaMa outputs, human-time measurements or occluded-observation identity recovery evidence.

S19.14 Official PartCATSeg checkpoint on the fixed PACO holdout

The official VOC-trained PartCATSeg checkpoint was chosen from the released canonical evaluation recipe before target inference. Source commit 4a9a88288659f49bc39ea4ee756dd45036552638, checkpoint SHA-256 51a0cb99254f3778a0d629e0053af937807e55b266ae0071438c141566a36766, adapters, inputs and evaluation rules were frozen before the 42-case run. This retrospective comparison reuses the original holdout crops, not a new untouched test. Both methods receive the same localized image pixels without true category/profile or reference masks; the historical HPID commands were audited for this information condition.

A single global vocabulary contains all 456 public PACO object-part categories, ordered by category ID, plus background. No per-case annotation or present-part list defines the vocabulary. The official unfiltered sem_seg_all output is argmaxed

after the official resize/postprocessing. Constant dummy targets only satisfy the forward API; changing them left scores unchanged in the VOC and PACO smoke controls. For memory limits, class-independent decoder layers use batches of 16 after full-vocabulary aggregation. On the VOC smoke image, unbatched versus batched decoding had identical argmax and maximum score difference 1.10e-6; this numerical control is not a universal bit-exact guarantee. Learned layers use float32 with TF32 and autocast disabled. Checkpoint loading had no missing or unexpected keys.

Each non-background raw label is split into every 8-connected component without area filtering or morphology. All components, including wrong-object and unmapped labels, enter geometric scoring. HPID retains native Parts and the released root/body rule; 27 root records are excluded in cases without an annotated body. PartCATSeg has no separate root record. Matching maximizes total mask IoU one-to-one before thresholding. The secondary semantic-union metric merges predictions by the fixed canonical alias bridge and averages IoU over annotated reference semantics, then over cases; it uses full crop masks without reference clipping. Wrong predicted object/domain labels are not repaired using the target category. This annotation-conditioned metric is not official dense-benchmark mIoU, and semantic label precision measures label presence rather than thresholded geometric precision.

All 42 outputs were valid and all cases were retained. The primary endpoint is case-macro geometric Part F1@0.25. Paired source-image percentile intervals use 10,000 draws, seed 20261009; secondary intervals are descriptive and unadjusted. Table S53 reports HPID-minus-PartCATSeg differences. The archived HPID baseline, native learned label maps, component counts, exact input/output hashes and scoring statistics passed independent audits.

TABLE S53. Learned-baseline transfer comparison on 42 original holdout crops

Endpoint	HPID	PartCATSeg	HPID − PartCATSeg	95% paired CI
Part F1@0.25 (primary)	0.3390	0.0311	+0.3078	[0.2404, 0.3746]
Part F1@0.50	0.1645	0.0066	+0.1579	[0.0983, 0.2258]
Part F1@0.75	0.0753	0.0000	+0.0753	[0.0302, 0.1329]
Part precision@0.25	0.3165	0.0173	+0.2992	[0.2305, 0.3673]
Part recall@0.25	0.4085	0.2435	+0.1650	[0.0596, 0.2608]
Semantic-union IoU	0.1905	0.0554	+0.1351	[0.0869, 0.1915]
Semantic-union recall@0.25	0.2816	0.0679	+0.2137	[0.1450, 0.2882]
Semantic label-presence precision	0.4575	0.1511	+0.3064	[0.2320, 0.3811]

The primary difference is +0.3078 [0.2404, 0.3746]; semantic-union IoU also favors HPID by +0.1351 [0.0869, 0.1915]. Interpretation is restricted to this VOC-to-PACO transfer, vocabulary and instance adapter. PartCATSeg produces 3,095 components (mean 73.69 per crop), including 1,328 of at most 16 pixels; fragmentation contributes to the precision burden. HPID averages 6.14 scored Parts. The fixed no-filter rule was not adjusted after results. These results do not establish a ranking on PartCATSeg's official dataset, general superiority over learned methods, or persistent-ID accuracy for a method without that interface. HOPS and KPS remain unexecuted.